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of environmentally
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growth: Comparing available
options and estimating
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study countries

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Measurement of environmentally sustainable productivity growth

Comparing available options and estimating trends for three
OECD case study countries

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Abstract

This report reviews six indices of environmentally sustainable productivity growth (SPG) that integrate agricultural total factor productivity (TFP) growth and trends in four agri-environmental externalities at the national level.

The findings reveal only minor differences across indices when environmental dimensions are assigned low weights, but these differences widen as weights increase. This variation reflects index-specific conditions under which SPG exceeds TFP. Some indices require externalities to grow more slowly than outputs or input (relative decoupling) while others require externalities to decrease (absolute decoupling).

Three case studies (Mexico, The Netherlands and New Zealand), illustrate potential drivers of SPG. In Mexico, improved market access resulting from trade agreements, alongside shifts in agricultural support appeared to be key factors. In the Netherlands, regulatory measures targeting ammonia, fertilisers and manure management contribute to positive SPG outcomes. In New Zealand, performance is associated with export-driven output growth, herd dynamics, and the adoption of new technologies.

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Key Messages

- This report compares six indices¹ of environmentally sustainable productivity growth (SPG²) that integrate agricultural total factor productivity (TFP) growth and trends in four agri-environmental externalities (emissions of greenhouse gases (CH₄, N₂O, CO₂), ammonia (NH₃) emissions and nitrogen and phosphorus run-off and leaching) at the national level.
- The results show that:
 - In the case of three of the reviewed indices (EATFPO, EATFPI, ESPII), SPG exceed traditional TFP growth when the selected environmental externalities grow more slowly than agricultural outputs or inputs (relative decoupling).
 - For the other three indices (ESPIO, WEPI, WEPO), SPG exceed TFP growth only if environmental externalities decline faster than increases in outputs (ESPIO) or in TFP (WEPI/WEPO) (absolute decoupling).
 - All SPG metrics provide useful information regarding the relative growth rates of and outputs and inputs and environmental externalities. However, positive SPG does not necessarily imply good environmental performance or declining environmental externalities.
 - Across all metrics, differences between SPG and TFP growth increase with the number of environmental dimensions and with the magnitude of their unit prices, reflecting the reduced relative weight of traditional TFP.
- Case studies for Mexico, the Netherlands and New Zealand illustrate how SPG, which incorporates environmental externalities, compares to traditional TFP growth estimates.
 - Estimated SPG generally exceeds TFP growth for three indices (EATFPO, EATFPI or ESPII) in the three countries, as they experienced relative decoupling of environmental impacts. However, for the other three indices (ESPIO, WEPI, WEPO), SPG is lower than TFP growth for Mexico and New Zealand. This can be explained by the simultaneous increases in environmental externalities and in agricultural outputs and inputs (i.e. no absolute decoupling).
- Factors explaining trends in agri-environmental performance are country-specific:
 - In Mexico, major shifts in agricultural support policies during the 1980s/90s, coupled with trade liberalisation and technology adoption in selected value chains, were key drivers of output growth. While externalities increased with this expansion, they did so at a slower pace.
 - In the Netherlands, declines in ammonia and nutrient inputs and balances during 1990-2010 largely explain the gap between SPG and TFP growth. Regulatory measures on manure, ammonia and fertiliser were instrumental in driving improved on-farm management and technology adoption.
 - In New Zealand, SPG dynamics reflect structural factors, shocks and technology adoption. In the 1990s, the rapid expansion in herd sizes was followed by a rapid increase in ammonia and greenhouse gases (CH₄, N₂O and CO₂). From the mid-2010s, as herd sizes stabilised and with the adoption of improved technologies and practices, the economic value of environmental externalities per unit of input and output began to decline.

Executive Summary

Achieving sustained output growth in an environmentally sustainable way is key to ensuring food security, while simultaneously providing livelihoods for farmers and improving the environmental sustainability of the sector. This requires producing more with less; achieving sustainable productivity growth is therefore essential for increasing output while reducing environmental pressures. Traditional productivity metrics, however, typically focus only on conventional outputs and inputs, leaving aside environmental externalities.

Several indices have been proposed in the literature to measure increases in agricultural productivity in ways that account for environmental dimensions. This report reviews alternative measures using six indices of environmentally sustainable productivity growth (SPG), by comparing their sensitivity to methodological choices. The six indices considered are: environmentally adjusted TFP indicators with input (EATFP-I) and output (EATFP-O) adjustments, the environmentally sustainable productivity index with input (ESPII) and output (ESPIO) adjustments, and the weak environment-productivity growth indicator where the weights of the externality are given by their importance relative to the value of inputs (WEPI) and outputs (WEPO).

Calculating an environmentally sustainable productivity index relies on several assumptions and choices, including the intended use of the index, the environmental dimensions included, and the perspective of the environmental valuation (producer vs. societal valuation). Each of these choices involves trade-offs between simplicity of estimation for improved interpretation versus more realistic representation with complex estimations. While many choices are technical, the perspective used for the valuation also depends on the intended use of the index and whether the index intends to capture sector-specific or broader social welfare impacts.

The analysis shows that SPG metrics differ in important ways. Some treat the environment as an output, others as an input, and they aggregate traditional and environmental outputs and inputs in different ways. As a result, growth rates derived from different indices vary, as do the conditions under which SPG rates exceed those of TFP.

- Most indices rely on the concept of an environmentally adjusted output (or traditional output) growing faster than traditional input (or environmentally adjusted input).
- Some indices (EATFPI, EATFPO, ESPII) require only relative decoupling between the output or inputs and externalities for their growth rates to exceed those of traditional TFP.
- Others (WEPI, WEPO, ESPIO) require absolute decoupling (i.e. a decrease in the overall level of externalities at a certain speed).

These differences stem from the way in which different methods aggregate environmental aspects and outputs or inputs.

Positive SPG rates do not necessarily imply a good environmental performance of the agricultural sector or that the production methods can be maintained over the long-term without degrading natural ecosystems. Instead, SPG metrics provide a valuable measure of economic performance that integrates the environment. While this report does not assess the environmental performance of agricultural production systems, it highlights the need for complementary analyses with indicators that also track the levels of environmental externalities.

As an illustration, SPG metrics and TFP growth rates were estimated under several different methodological assumptions for Mexico, the Netherlands and New Zealand. These estimations were conducted from 1990 onwards, using agri-environmental indicators representing four environmental externalities (direct emissions of greenhouse gases CH₄, N₂O and CO₂, ammonia emissions, and estimates of nitrogen leaching and phosphorus run-off). Two insights emerge:

- First, when weights assigned to the environment are low relative to those assigned to traditional outputs/inputs, differences across the indices are small. This is intuitive as traditional TFP drives these results in these cases.
- Second, as the weight assigned to the environmental externalities increases, differences between indices become more visible. In countries where relative (but not absolute) decoupling of externalities from output/input has occurred (e.g. Mexico and New Zealand), SPG rates are generally higher than TFP growth for 3 out of 6 indices. For the Netherlands, where the reduction of externalities has been faster than output and input growth since the 1990s, SPG rates are consistently higher than traditional TFP.

Further country-level insights can be drawn by disaggregating trends in SPG metrics.

- Components explaining SPG differ across countries: whereas environmental performance (and ammonia emissions in particular) contributes the most towards SPG in the Netherlands, the growth in traditional outputs and inputs remain the key components explaining SPG in the cases of Mexico and New Zealand, with adjustments being relatively small.
- Drivers of these components are country-specific: in Mexico, improved market access as a result of trade agreements, alongside shifts in agricultural support and output expansion are considered key factors. In the Netherlands, regulatory factors (especially targeting ammonia, fertilisers and manure) are considered key drivers of positive adjustments to SPG. In New Zealand, export-driven output expansion, herd dynamics, and technology or practice adoption are seen as important explanations for the observed performance over time.

1 Introduction

Environmentally sustainable productivity growth – ensuring that output increases are achieved in environmentally sustainable way – is a key element in addressing the triple challenge of food systems (OECD, 2021^[1]). Globally, total factor productivity (TFP) growth remains the main driver of output growth since the 1990s (Fuglie, Morgan and Jelliffe, 2024^[2]). At the same time, agriculture continues to have a substantial environmental footprint, accounting for approximately 70% of global freshwater demand (Gruère, Shigemitsu and Crawford, 2020^[3]), occupying nearly 40% of terrestrial land (OECD, 2019^[4]) and accounting for approximately 14% of emissions of greenhouse gases (GHG),³ consisting of CO₂, CH₄, N₂O (Tubiello et al., 2021^[5]; FAO, 2024^[6]).

Despite gains in productivity and output growth, these improvements have not always been mirrored in environmental outcomes, highlighting a growing divergence between agricultural performance and agri-environmental indicators. For the OECD, output metrics have consistently improved since 1990 irrespective of the metric used (calories or value of output). However, the same does not apply for agri-environmental indicators, where progress on several agri-environmental indicators such as nutrient balances or emissions of GHG was largely concentrated in the 1990-2010 period (OECD, 2025^[7]). Since 2010 progress has slowed or, in some cases, reversed (e.g. GHG emissions, nutrient balances per hectare). Beyond these indicators, trends in the Farmland Birds Index, used to proxy farmland biodiversity, have also been negative for the majority of OECD countries over the last decade (OECD, 2025^[7]).

Monitoring improvements across both dimensions underscores the need to use metrics capable of capturing both production and environmental dimensions. By combining information on agricultural production and agri-environmental performance, measures of environmentally adjusted productivity growth offer one potential way forward (OECD, 2024^[8]). Integrating environmental dimensions into productivity metrics, however, remains complex.

The challenge of constructing environmentally sustainable productivity measurements is not confined to the agricultural sector. Beyond the agricultural sector, economy-wide metrics, such as the Environmentally adjusted multifactor productivity (EAMFP), already exist (Cárdenas Rodríguez et al., 2023^[9]; Cárdenas Rodríguez, Hašič and Souchier, 2018^[10]). In the agricultural sector, multiple approaches have been proposed to measure environmentally sustainable productivity growth (SPG), as summarised in recent publications (OECD, 2022^[11]; Bureau and Antón, 2022^[12]). Differences in the metric used, the environmental dimensions considered, and the valuation of environmental externalities make direct comparison across different SPG metrics challenging.

The aim of this paper is three-fold. First, it aims to highlight several methodological choices when developing an SPG metric applied to the agricultural sector. This includes the choice of the index, the choice of the environmental dimensions to be included and the value (weight) assigned to the environmental externalities. Second, using three OECD countries as case studies (Mexico, the Netherlands, and New Zealand⁴), the sensitivity of SPG rates to these different choices (choice of index, weights, and aggregation method) is tested. Finally, the paper then interprets the country-specific SPG rates considering the country-specific performance across various dimensions of agri-environmental performance and proposes different alternatives to visualise the results.

2 Overview of several existing environmentally adjusted productivity metrics

Developing a SPG metric requires several important methodological choices

Developing an environmentally adjusted productivity metric requires several key methodological choices.⁵ This paper discusses three particularly important ones: a) the choice of index, b) the selection of environmental dimensions to include; and c) the approach used to value environmental externalities in constructive SPG estimates.

The first choice relates to the index which is used to coherently aggregate the traditional productivity components (outputs and inputs) and the environmental dimensions. Several ways have been proposed to measure SPG. This paper focuses on Environmentally adjusted TFP indices using a growth accounting framework (EATFP), the Environmentally Sustainable Productivity Index (ESPI), as well as the Weak Environmental Performance (WEP) metric. All indicators are explained in the next sub-section⁶.

The second key methodological choice concerns the environmental dimensions included in the SPG metric. Previous research has incorporated environmental dimensions into productivity measures in different ways. Some studies have adjusted productivity metrics for weather shocks, whereas others have focused on accounting for single or multiple environmental externalities (Baldoni et al., 2025^[13]; Ang, Dakpo and Pieralli, 2022^[14]; Gu, Hussain and Willox, 2019^[15]; Chancellor et al., 2021^[16]; Nanere et al., 2007^[17]; Fuglie et al., 2020^[18]; Färe et al., 1993^[19]; OECD, 2022^[11]). Importantly, although environmentally adjusted metrics can, in principle, incorporate both positive and negative externalities, most existing research concentrates on negative externalities.

The third choice relates to the valuation of the environmental variables and the selection of weights associated with the environmental dimension. Constructing an SPG metric requires determining the relative weight of the environmental dimension(s) vis-à-vis production. Importantly, while this is, to a large extent, a methodological choice, this choice has profound effects on the impact of the included environmental dimensions on the growth rates. Beyond this, it can also lead to a different interpretation of the metrics. Producer-centric valuations of the environment, while arguably more methodologically consistent,⁷ disregard off-farm impacts of the externalities (e.g. on health, biodiversity). Valuations that go beyond farm abatement costs capture broader costs associated with the externality. However, it is often argued that they do so at the expense of consistency with standard production theory. From a technical perspective, while producer-centric valuations remain more common in the literature, different authors have opted for different methodological choices.

Existing evidence suggests that methodological choices may have a sizeable effect on the estimated SPG rates. A recent study focusing on the agricultural sector in EU Member states tested the sensitivity of SPG metrics to alternative methodological choices, with a focus on index selection and weights (Baldoni et al.,

2025_[13]). The authors identify several factors to be important in determining how different SPG rates will be from traditional TFP growth rates, including the choice of index, the unit value assigned to the environmental externality, and whether externalities are included by adjusting the input or output.

Methodological choices also have two additional implications. First, the perspective used to value environmental externalities— whether based on producer or societal valuations— changes what is being measured, ranging from a proxy of producer profitability growth to an indicator that captures the value and costs of production from a societal perspective, which is referred to by some authors as the profitability of production from society's perspective (Henningsen, 2025_[20]). Aligning the valuation perspective with the intended use of the index is therefore essential. Second, methodological choices related to environmental valuation may constrain index selection. When environmental externalities are large relative to output values, output-adjusted SPG measures, such as those used by Baldoni et al. (2025_[13]) can theoretically result in negative adjusted outputs. In such cases, particularly under high valuations of externalities, input-adjusted indices have desirable properties as they avoid specifying negative adjusted outputs.

A summary of several indices proposed in the literature

Total Factor Productivity TFP

Total Factor Productivity (TFP) is widely used as an aggregate measure of economic performance in the agricultural sector (OECD, 2022_[11]) and has been the main driver of output growth since the 1990s (Fuglie, Morgan and Jelliffe, 2024_[2]). Improvements in agricultural TFP are an important component of sustainable growth in the agricultural sector as they reflect the capacity to produce greater quantities of outputs with fewer inputs, which is important to preserve natural resources.

Mathematically, TFP is defined by a ratio of aggregate output to aggregate inputs:

$$TFP_t = \frac{YI_t}{XI_t} \quad (1)$$

Since agriculture is a multi-output multi-input activity and YI_t and XI_t are often expressed as quantity indices where the relative importance of the different outputs and inputs are given by revenue and cost shares. In the case of the FAO output index used by USDA, it is given by $YI_t = \frac{\sum_i p_{y,i}^0 q_i^t}{\sum_i p_{y,i}^0 q_i^0}$, where the output index is given by the aggregate value of production (sum of the production of each commodity i at time t (q_i^t), valued at a time-invariant base year price for a given commodity ($p_{y,i}^0$)) divided by the aggregate value of production in the base year (defined as the sum of the production of each commodity i at the base period (q_i^0) valued at a time-invariant base year price ($p_{y,i}^0$)).

While TFP growth implies that aggregate output is growing faster than aggregate input, it does not provide any information on potential environmental consequences of output growth (Bureau and Antón, 2022_[12]). This has motivated the adjustment of TFP metrics to include environmental externalities and several indices have been proposed. This paper focuses on five indices⁸ of environmentally adjusted productivity. For the conceptual exposition below, time-invariant prices are assumed.⁹

Environmentally Adjusted Total Factor Productivity (Output) [EATFPO]

The first metric considered is the Environmentally adjusted TFP with output adjustment (EATFPO). This metric requires constructing an environmentally adjusted output (by aggregating together traditional agricultural outputs and externalities). It measures the ratio of the environmentally adjusted output to traditional input. Mathematically, this metric generally takes the following form:

$$EATFPO_t = \frac{EAOI_t}{XI_t} \quad (2)$$

Where the numerator ($EAOI_t$) is an environmentally adjusted output index over aggregate input index (XI_t). A common way of adjusting the output is by assigning a monetary value to the externalities and subtracting the estimated monetary value of negative externalities¹⁰ ($p_e * E_t$ in the case of one externality) from the total monetary value of traditional output ($Y_t = \sum_i p y_i^0 q_i^t$) (Bureau and Antón, 2022^[12]). If this approach is used and if time-invariant prices are assumed, $EATFPO$ ¹¹ in time t relative to a base year can be expressed as:

$$EATFPO = \frac{\frac{EAO_t}{EAO_0}}{\frac{\sum_j p x_j^0 X_j^t}{\sum_j p x_j^0 X_j^0}} = \frac{\frac{\sum_i p_{y,i}^0 q_i^t - p_e E^t}{\sum_i p_{y,i}^0 q_i^0 - p_e E^0}}{\frac{\sum_j p_{x,j}^0 X_j^t}{\sum_j p_{x,j}^0 X_j^0}} \quad (3)$$

Conceptually, treating externalities as bad outputs is more intuitive since externalities (at least the ones considered in this paper) are generally by-products of externality-causing intended outputs (Murty, Robert Russell and Levkoff, 2012^[21]). However, mathematically and from a practical perspective, treating externalities as bad outputs also has disadvantages. Specifically, conceptually, it can result in low or negative adjusted outputs (when costs of externalities exceed value of production). In such situations, this can limit the range of plausible shadow values¹² (to avoid negative net outputs)¹³ and it can also potentially lead to more volatile growth rates.

Environmentally Adjusted Total Factor Productivity (Input) [EATFPI]

The second index considered is the Environmentally-adjusted TFP with an input adjustment [EATFPI] – In this measure, the adjustment is done by constructing an environmentally-adjusted input measure. This is done by augmenting the input (the denominator) in such a way that it includes both traditional inputs (typically land, labour, capital, feed and fertiliser in the case of the USDA ERS International Agricultural Productivity database) and environmental externalities. In this case, conceptually, externalities can be seen as proxies for consumption of natural resources which are treated as an additional input. This metric therefore takes the ratio of traditional output and environmentally-adjusted input index ($EAII_t$) and can be expressed as:

$$EATFPI_t = \frac{YI_t}{EAII_t} \quad (4)$$

If prices for both inputs and externalities can be obtained equation (4) can be expressed as:

$$EATFPI_t = \frac{\frac{\sum_i p_{y,i}^0 q_i^t}{\sum_i p_{y,i}^0 q_i^0}}{\frac{\frac{EAI_t}{EAI_0}}{\frac{\sum_j p_{x,j}^0 X_j^t + p_e E^t}{\sum_j p_{x,j}^0 X_j^0 + p_e E^0}}} = \frac{\frac{\sum_i p_{y,i}^0 q_i^t}{\sum_i p_{y,i}^0 q_i^0}}{\frac{\sum_j p_{x,j}^0 X_j^t + p_e E^t}{\sum_j p_{x,j}^0 X_j^0 + p_e E^0}} \quad (5)$$

Adjusting traditional TFP by extending the inputs is conceptually less straightforward as it treats environmental externalities as consumption of the environment. However, mathematically, it has convenient properties, including the fact that the denominator is never negative, does not impose a limit on the shadow prices, and, in cases where the values of externalities are large relative to output and inputs and change quickly, growth rates are likely to be less sensitive.

Environmentally Sustainable Productivity Index [ESPI-O]

The third index considered in this paper is the Environmentally Sustainable Productivity Index [ESPIO] – The ESPI is an index-value based approach which is closely related to work on sustainable productivity indices (O'Donnell, 2018^[22]; O'Donnell, 2022^[23]; Njuki and O'Donnell, 2025^[24]). The environmentally sustainable productivity index is defined as the ratio of an environmentally adjusted output index over an input index (Cobourn et al., 2024^[25]). Mathematically, it is expressed as:

$$ESPIO_t = \frac{EAOI_t}{XI_t} = \frac{(YI_t)^{1-\alpha}(EI_t)^{-\alpha}}{XI_t} \quad (6)$$

Importantly, however, while the ESPI also relies on the concept of an environmentally adjusted output, there are important differences with the EATFPO metric. First, all quantity indices (YI_t , EI_t , XI_t) are expressed against a common base-year value¹⁴ for all countries which enables cross-country comparisons of the levels of ESPI (Cobourn et al., 2024^[25]). Second, there is an important difference in terms of how the Environmentally adjusted output is constructed. Whereas EATFPO metrics usually rely on prices to aggregate market and non-market outputs, the ESPI uses a parameter α which captures the relative importance of traditional output and bad outputs (i.e. externalities). As a result, while the relative importance of the bad outputs in the EATFP metric depends on their calculated monetary value (which implies that the weight of the value of the environmental externality in the environmentally adjusted output will be country-specific), the ESPI assumes this is defined by a alpha, which is constant across all countries in the original paper (Cobourn et al., 2024^[25]).

Environmentally Sustainable Productivity Index with input adjustment [ESPII]

The fourth index considered in this paper is a modified version of the Environmentally Sustainable Productivity Index [ESPII] where the adjustment is done via the inputs rather than outputs. While this index does not feature in the contribution that proposes ESPI (Cobourn et al., 2024^[25]), it is calculated in this paper to facilitate the comparison with SPG metrics that adjust inputs rather than outputs (i.e. EATFPI). Mathematically, adjusting ESPI through the denominator can mathematically be expressed as

$$ESPII_t = \frac{YI_t}{EAII_t} = \frac{YI_t}{(XI_t)^{1-\alpha}(EI_t)^{\alpha}} \quad (7)$$

Where the parameter α , bounded between 0 and 1¹⁵, now expresses the relative importance of the externalities vis-à-vis traditional inputs, rather than outputs (as in ESPIO). There are two reasons for introducing this index. The first is that it is more comparable to the EATFPI since both treat externalities in a similar fashion. The second reason is that it has a slightly different logic to ESPIO which results in different conditions (more in-line with EATFPI) as to when ESPII exceeds TFP. Specifically, while the evolution in the nominator of the ESPIO essentially looks at whether improvements in environmental performance keep up with improvements in production (i.e. for a positive growth in output it requires a reduction in the externalities at a specific pace for EAOI to exceed YI), ESPII asks whether a country is able to keep the growth in EI lower than that of XI (i.e. consumption of the environment below the increase in use of inputs). As will be seen in subsequent sections, this has implications in terms of how the metrics behave relative to traditional TFP.

Weak Environment-Productivity Growth [WEP]

The final metric considered is the weak environment-productivity (WEP) growth. The underlying idea of the WEP is to allow full substitution between an environmental outcome and TFP growth (Henderson and Lankoski, 2023^[26]). The way the WEP is constructed is by adjusting TFP growth rates by calculating a weighted average of traditional TFP growth and growth of environmental metric(s)¹⁶ (Henderson and Lankoski, 2023^[26]). Mathematically, it is expressed as:

$$g(WEP) = (1 - \alpha) * g(TFP) - \alpha * g(E) \quad (8)$$

Where the parameter α denotes the relative weight given to the growth of environmental externalities (emission intensity of GHG emissions in the original paper) and TFP. One important difference between WEP and the four other metrics outlined above is that, while all other metrics adjust either the output or input, the WEP uses a weighted average of unadjusted TFP growth and growth of environmental externalities. In the case of WEP, two versions of the WEP will be used, namely WEPI and WEPO. The only difference between these two is the value of α . In short, two values of α are considered, one that is based on the relative *importance* of externalities vis-à-vis inputs (WEPI) and another vis-à-vis outputs (WEPO). Values used are provided in Table A A.1 in the appendix. While these are, strictly speaking, not different indices (the only difference is the weights assigned to the growth of the externalities), these will be referred to separately in the results section.

Interpretation: How do different SPG indices compare?

By taking a ratio of aggregate output to aggregate inputs, TFP can be intuitively interpreted as the capacity to produce “more with less” (i.e. more traditional output with less traditional inputs). TFP growth rate also has an intuitive interpretation in that it is the difference between the growth rate of aggregate output and aggregate input (i.e. the growth in output that cannot be attributed to the growth in inputs). A positive growth rate over time therefore implies that aggregate output is increasing faster than aggregate input.

The interpretation of environmentally adjusted productivity metrics is more complex and index-specific. However, for indices that adjust output and inputs, growth rates are positive when environmentally-adjusted output growth exceeds input growth (in the case of output adjustment) or when output growth exceeds environmentally-adjusted input growth (in the case of input adjustment). As such, environmentally adjusted productivity metrics still capture the spirit of “producing more with less”. However, as will be shown later, the interpretation can differ and the conditions under which the growth rate of SPG metrics exceeds those of traditional TFP differ depending on the metric.

The different indices described in the previous section all share several important features.

All reviewed SPG indices use “weights” to aggregate output and externalities and converge towards TFP when weights are 0

An important common feature across all reviewed metrics is the reliance on explicit or implicit valuation of the environment. All metrics rely on some form of aggregation between environmental and non-environmental outputs/inputs that relies on weights. Such weights can be based on externality prices, explicit parameters (e.g. alpha coefficient in ESPI). If time-invariant prices are used and denoting the importance of the environment as γ ¹⁷, this implies that metrics in equations (3), (5) and (6)-(8) can be represented in index form as a function of a parameter γ (Table 1 and derivations are provided in Annex A).

This has two implications. First, all SPG metrics described tend to traditional TFP as the weight assigned to the environmental variables approaches 0 (i.e. as p_e tends to 0 in equations (3) and (5) or as α tends to 0 in equations (6)-(8)). This implies that differences in growth rates across the different indices will increase as the weight of the environment increases since the relative weight assigned to factors captured in traditional TFP will be lower. Second, it is possible to calculate gamma in a way that is conceptually similar across indices (i.e. set *gamma* to reflect the *gamma* observed in EATFPI or EATFPO) such that *gamma* has the same interpretation across all metrics.¹⁸

Table 1. Expressing different indices as a function of a parameter gamma

Metric	Expression
EATFPO	$\frac{EAOI_t}{XI_t} = \frac{W * ((1 - \gamma) * YI_t - \gamma * EI_t)}{XI_t} ;$ $\text{with } W = \frac{p_y Y_0 + p_e E_0}{p_y Y_0 - p_e E_0} ; \gamma = \frac{p_e E_0}{p_y Y_0 + p_e E_0} ;$
EATFPI	$\frac{YI_t}{EAII_t} = \frac{YI_t}{(1 - \gamma)XI_t + \gamma * EI_t} ;$ $\text{with } \gamma = \frac{p_e E_0}{p_x X_0 + p_e E_0}$
ESPIO	$\frac{(YI_t)^{1-\gamma}(EI_t)^{-\gamma}}{XI_t} ; \text{ where } \gamma = \alpha ; 0 < \gamma < 1$
ESPII	$\frac{YI_t}{(XI_t)^{1-\gamma}(EI_t)^{\gamma}} ; \text{ where } \gamma = \alpha ; 0 < \gamma < 1$
WEP	$g(WEP) = (1 - \gamma) * g(TFP) - \gamma * g(EI) ; \text{ where } \gamma = \alpha ; 0 < \gamma < 1$

Note: For simplicity of exposition, in Annex A and in this table the focus is on a case with one single output (Y), one single externality (E) and one single input (X). For ease of notation, superscript 0 was removed from all prices, but all prices in the table are time-invariant. In other words, p_y , p_e and p_x are equivalent to p_y^0 , p_e^0 , and p_x^0 .

Despite similarities, the nature and interpretation of the weight differs across indices

One important difference relates to how the importance of the environment (γ) is determined across indices (Table 1). In EATFP metrics, γ has an intuitive economic interpretation but requires the user to know/estimate the unit price of the externalities (the ratio of the value of the externalities relative to the total joint economic value of outputs/inputs and externalities, which is likely to be country-specific). ESPI and WEP, on the other hand, have imposed the same weight across all countries (though this need not be the case), but provide greater flexibility in the way that γ is set by allowing it to vary between 0-1. When γ is set arbitrarily the more intuitive economic interpretation is lost, but the added flexibility also means that γ can be set in a way that preserves an intuitive interpretation. It represents the sensitivity of the policy maker or the society towards environmental issues and can be interpreted as the “weight” given to environmental externalities compared to economic outcomes that can be measured at market prices. If the user so decides, it can be set in an analogous way to gamma in EATFP metrics. This is the approach followed in the paper, to ensure a greater degree of comparability and to preserve the intuitive interpretation of *gamma* from EATFP metrics.

SPG is larger than TFP growth when output/input growth is sufficiently decoupled from environmental externalities growth

The conditions for an environmentally adjusted productivity growth (EAPG) to be positive are quite similar across metrics (except for WEP). SPG rates will be positive whenever the growth of environmentally adjusted output exceeds that of traditional input (in the case of EATFPO and ESPIO) or when output growth exceeds that of environmentally adjusted input growth (in the case of EATFPI and ESPII) (Column (2) of Table 2). In the case of WEP, the interpretation is slightly different as it requires that the weighted growth rate of TFP compensates the weighted growth rate of the externality.¹⁹

Another important commonality across the different metrics is that for SPG rates to exceed TFP growth rates always requires a sufficient degree of decoupling between outputs/inputs and externalities. The definition of “sufficient” level of decoupling, however, differs across indices (Column (3) of Table 2). In this sense, indices can broadly be classified in two types, depending on the degree of decoupling required. Specifically:

- **Some indices require relative decoupling between the externality and output/input** (EATFPO, EATFPI, ESPII) – For these indices, for the SPG rate to exceed the growth rate of TFP, the growth rate of the externalities needs to be slower than that of output (EATFPO) or input (ESPII, EATFPI). These metrics therefore require relative decoupling between the considered environmental externalities and output or input for the growth rates of these metrics to exceed those of TFP.
- **Other SPG metrics require absolute decoupling between output/TFP growth and externality growth** (ESPIO, WEP) - In the cases of WEP and ESPIO, for the growth rate of these metrics to exceed TFP growth, they require a reduction in environmental externalities at a pace faster than that of output growth. In other words, assuming that output increases, these indices require not only absolute decoupling (negative growth of environmental externalities) but that the externalities decrease at least as fast as output (ESPIO) or TFP (WEP) increases. Only when output growth is negative do these indices allow for growth in environmental externalities to be compatible with higher levels of SPG than TFP growth. This therefore implies that, for growing levels of output/TFP (which is the general case), these two metrics require stronger environmental performance for their growth rates to exceed that of TFP when compared to the other three.

Table 2. When are SPG rates positive? When does it exceed TFP growth?

(1) Metric	(2) SPG>0 when	(3) SPG>TFPG when
EATFPO	$g(EAOI) > g(XI)$	$g(EI) < g(YI)$
EATFPI	$g(YI) > g(EAII)$	$g(EI) < g(XI)$
ESPIO	$g((YI_t)^{1-\gamma}(EI_t)^{-\gamma}) > g(XI)$	$g(EI) < -g(YI)$
ESPII	$g(YI) > g((XI_t)^{1-\gamma}(EI_t)^{\gamma})$	$g(EI) < g(XI)$
WEP	$(1 - \gamma) * g(TFP) > \gamma * g(EI)$	$g(EI) < -g(TFP)$

Note: $g(\cdot)$ denotes the growth rate. As such $g(YI)$ is the growth rate of YI (i.e. output index). EAOI stands for Environmentally adjusted output index. EAII stands for environmentally adjusted input index. YI, EI and XI denote aggregate output, externality, and input indices, respectively. Derivations of the results are provided in Annex B.

Choice of dimensions: What is included and how has practical consequences

When choosing the environmental dimensions, a decision needs to be made regarding which externalities to consider and in what form to consider them. Conceptually, both positive and negative externalities can be included in an environmentally adjusted metric. Ideally, both types of externalities would be included as agriculture delivers both agri-environmental services (e.g. habitat provision, visual amenities, care farming) and is also associated with negative environmental externalities (Pretty et al., 2000^[27]). There are several examples of attempts to value both positive and negative externalities in agriculture with examples including Canada and the Netherlands (Jongeneel, Polman and Cornelis Van Kooten, 2016^[28]; Skolrud et al., 2020^[29]) which have all found the economic value of positive externalities to be substantially lower than those of negative externalities. However, when calculating SPG metrics for multiple countries, a focus on negative externalities is more common, not least because of a lack of comparable data sources on quantities and values of positive externalities across countries. This makes the inclusion of positive externalities more challenging and most attempts to propose EAP metrics at the national level have focused on the inclusion of negative externalities.

Even when considering only negative externalities, the choice of the externalities to include is likely to differ based on the purpose of the exercise. There are examples in the literature that focus on the inclusion of different types of externalities, including GHG emissions (Ang, Dakpo and Pieralli, 2022^[14]; Henderson and Lankoski, 2023^[26]), water (Vrachioli and Tzouvelekas, 2022^[30]; Fuglie et al., 2020^[31]), soil erosion (Nanere et al., 2007^[32]), pesticides (Ball et al., 2001^[33]; Wang, Hussain and Yan, 2024^[34]) as well as several attempts to include several dimensions simultaneously (Nin-Pratt, 2025^[35]; Ball et al., 2001^[33]; Cobourn et al., 2024^[25]). Below, four important aspects related to the inclusion of externalities are highlighted.

One fundamental constraining factor influencing the choice of externalities is data availability. For analyses based on the OECD agri-environmental indicator database, for instance, data gaps vary by indicator. There are fewer data gaps for indicators related to GHG emissions, nutrient balances, and, to some extent, ammonia emissions. On the other hand, data gaps are much larger for indicators related to domains such as biodiversity, water and erosion (Cobourn et al., 2024^[25]; Henderson and Lankoski, 2023^[26]). Another potential aspect that can inform the choice of dimensions beyond data availability relates to the relative importance of the dimensions being considered in monetary terms (i.e. the overall economic importance of the externality). Across valuation studies carried out in the Netherlands, Canada and the UK, GHG emissions, ammonia emissions and metrics related to nutrients consistently ranked as important in terms of their overall economic value (Pretty et al., 2000^[27]; Skolrud et al., 2020^[29]; Jongeneel, Polman and Cornelis Van Kooten, 2016^[28]).²⁰

Another second important aspect relates to the definition used for specific externalities. In the case of GHG emissions at the national level, for instance, the definition used (e.g. including/excluding Land use, land use change and forestry [LULUCF]) as well as the methodology used to estimate the level of emissions (e.g. Tier 1 vs. Tier 2 data) can have different implications in terms of the results. Different authors have chosen different definitions of GHG emissions when adjusting these metrics. Some authors have focused on narrower definitions of GHG emissions from agriculture (Cobourn et al., 2024^[25]), whereas other authors have adopted broader definitions that include additional components, such as net forest conversion (Ang, Dakpo and Pieralli, 2022^[14]). While the choice is very clear for farm-level analysis, arguments can be made for both choices when considering national-level analyses. This matters for two reasons. First, the two different definitions (including or excluding LULUCF) may imply different total quantities (and hence different economic value of the externalities, especially if using external shadow prices). Second, in some cases (e.g. Australia), trends in total quantities between the two definitions may be very different.

A third aspect that is important is to avoid overlap between dimensions. For some environmental externalities there may be an overlap. For instance, since part of the total nutrient inputs contributes to GHG emissions, it may be problematic to include both without adjusting nutrient indicators. It is therefore important to avoid double-counting of certain externalities.

Finally, not all externalities may be relevant for every country or sector within a country. It is thus important to have clarity on the purpose of the index. In instances where indices are expected to be used solely for temporal comparisons within a country/sector, this could justify the inclusion of a set of externalities most relevant to the country. In other contexts, when the possibility of cross-country comparison is an intended objective, it may be important to consider sets of externalities relevant to the group of countries as a whole, even if some externalities may not be relevant for one individual country/sector.

This analysis focuses on four negative externalities: agricultural GHG emissions (excluding LULUCF), ammonia emissions and estimated leaching and run-off calculated as a share of the total nutrient balance. An in-depth explanation of the definitions and approach used is provided in Annex C.

Choices related to environmental valuation: Different approaches to the valuation of the environment will impact results and change the interpretation of the SPG metrics

Environmental externalities can be valued in different ways, and this can introduce substantial uncertainty around the values attributed to the environment as well as create challenges in terms of methodological consistency. Different authors have approached the question of environmental valuation differently (Bureau and Antón, 2022^[12]). Most studies rely on shadow prices, either estimated endogenous prices (i.e. prices estimated using statistical techniques directly from the data) or use external prices²¹ (e.g. abatement costs, estimated damages from the externality) from the literature.²² One important aspect of using shadow prices is that they enable the combination of environmental and productivity metrics, which means that they at least capture environmental considerations to some extent. Different shadow prices have different strengths and weaknesses value improvements in environmental performance differently.

Estimated shadow prices are more methodologically consistent, but are sensitive to methodological choices and can preclude the inclusion of important externalities

Estimated endogenous prices (i.e. shadow prices) are more theoretically consistent assuming that opportunity costs of abatement activities can be modelled as foregone production of output (Bureau and Antón, 2022^[12]). Relative prices between environmental externalities stemming from such an approach are more comparable than if these are sourced from difference sources. Popular approaches to estimate shadow prices in this way include regression analysis, stochastic frontier analysis (SFA) and data envelopment analysis (DEA).

However, the econometric estimation of shadow prices faces several challenges. First, estimates are generally sensitive to the choice of estimation approach and the sample used (Luu and de Menna, 2025^[36]; Bureau and Antón, 2022^[12]). Second, estimated shadow prices are also likely to be sensitive to the definition of the externality (e.g. on-farm vs. AFOLU GHG emissions²³) and there are also challenges related to econometric estimation of shadow prices for externalities whose values are only observed at irregular intervals (e.g. erosion). Third, econometric estimation of shadow prices can condition the dimensions that can be included in an environmentally adjusted productivity metric. In cases where estimated shadow prices are negative or statistically insignificant, a given dimension may not be included, even when it is of policy relevance. For instance, in the case of the estimation of environmentally adjusted MFP for OECD countries, authors do not account for five pollutants as coefficients were statistically insignificant (Cárdenas Rodríguez, Haščíč and Souchier, 2018^[10]). This can occur for a variety of reasons, including estimation issues, lack of internalisation of the externality, or there simply being no trade-off between the externality and output. While this paper focuses only on environmental sustainability and does not consider social dimensions of sustainability, the issue of the lack of a statistically significant relationship

between output and the externality is likely to also be important when considering the social dimension of sustainability where trade-offs may be harder to detect econometrically.

Beyond estimation challenges, endogenous shadow prices face two important limitations. First, they focus on marginal changes, with their magnitude reflecting the cost to a farmer of reducing a given externality — typically in terms of foregone revenue or profit. The focus on marginal changes implies that they often do not capture features such as tipping points and do not provide information on whether the level of externalities is sustainable or not. Second, when producer abatement costs and societal damages are misaligned—for example, with low abatement cost and high damage— endogenous shadow prices may not adequately reflect the harm they cause. In the absence of an incentive framework that conveys these signals to farmers, the weights assigned to the externalities may understate their societal impact.

External shadow prices provide more flexibility, allow the inclusion of off-farm damages of externalities, but generally imply a lower degree of methodological consistency

An alternative to using estimated prices is to use external shadow prices from the literature (termed external in this report) (Bureau and Antón, 2022^[12]). This can be done, for instance using market approximations of the abatement costs (e.g. prices in emission trading schemes). Such an approach values the externality from the perspective of a producer. However, they do not capture the damage to society as a whole (Bureau and Antón, 2022^[12]), which would imply using valuations of the environment that are based on factors that extend beyond the farm (i.e. in this paper, this is referred to as from the perspective of society). If societal damages exceed by a large amount of private abatement costs, then using estimated shadow prices (or external shadow prices using a producer perspective) underestimate the value of the externality to society (Henningsen, 2025^[20]). Using external shadow prices provides added flexibility for policymakers to capture societal damages (if this is the intended purpose) and can help ensure that every dimension has positive prices²⁴ and can be included. External shadow prices may also be more intuitive to understand for policymakers.

However, shadow prices from external sources are often sourced from different sources that use different methodologies and are often associated with substantial levels of uncertainty, which can create methodological inconsistencies for two reasons. First, it may distort the relative prices of the different environmental externalities.²⁵ Second, when considering a societal perspective, external shadow prices can also create an inconsistency with the inputs and outputs (which are valued using the perspective of the producer). The degree of the inconsistency depends on the degree to which input costs and output at market prices approximate their social value.

The choice of environmental valuation can affect the interpretation of the metric

The chosen environmental valuation can also have implications on the interpretation of SPG. While many aspects of the valuation of externalities are technical questions (e.g. the choice of estimator), the perspective used to value the environment also depends on what is intended to be captured by the environmentally adjusted metric. Different approaches have been used in the literature. Valuation from a producer perspective is more common and more methodologically consistent since outputs and inputs are also valued from the same perspective.²⁶ However, whenever societal damages of the externalities far exceed abatement costs, weights assigned to the externalities are too low from a society's perspective²⁷ (Henningsen, 2025^[20]) and that using producer-centric valuations would lead to a lower weight assigned to the environment. In terms of interpretation, shifting from a producer-centric valuation to a society-centric valuation means that the focus shifts from the evolution of private profitability (i.e. the trends in changes of output and inputs valued from the perspective of the producer) to profitability of production from a societal perspective (i.e. societal valuation of outputs and costs of production).

3

Comparison of the growth rates across different SPG metrics: a case study approach

Main results of the case study analysis

This section presents the average growth rates obtained when calculating the growth rates using the different metrics, environmental dimensions, and values for the environmental externalities. An overview of the limitations of the analysis is provided in Section 3.3.

With regards to the data used, value of output data is sourced from FAOSTAT and includes crop and livestock output but excludes aquaculture. The input index is sourced directly from the USDA International Agricultural Productivity data product (U.S. Department of Agriculture, 2024^[37]). With regards to the choices of externalities, the focus is on ammonia emissions, GHG emissions from agriculture (excluding LULUCF), and estimated run-off and leaching for nitrogen and phosphorus. For ammonia and GHG emissions, data are sourced directly from the Agri-environmental database of the OECD (OECD, 2025^[38]). For nitrogen and phosphorus, an estimate of run-off and leaching is computed by combining nitrogen and phosphorus inputs from the OECD Agri-environmental database with estimates of the fraction of run-off and leaching. Estimates of run-off and leaching come from national inventory reports and national sources (in the case of nitrogen), or global modelling assumptions and/or national sources (in the case of phosphorous). A list of the variables used, their sources and assumptions used for the estimates are available in Annex C (Table A C.1, Table A C.2).

To aggregate the environmental variables, outputs and inputs, valuations of the different environmental externalities are used. For producer-centric valuations, most external shadow prices are sourced from either country-specific or global studies that analyse abatement costs/alternatives. For societal valuations, the paper relies on already-existing databases that provide ranges of damages of the included externalities (De Vries et al., 2024^[39]; Lord, 2023^[40]). However, in some cases (e.g. phosphorous in New Zealand), national sources that provide a proxy of replacement costs are used. The list of prices used is summarised in Table A C.3. In cases where indices depend on an exogenous parameter (e.g. ESPI), these have been calculated in a way that replicates the logic followed EATFPO (for ESPIO) and EATFPI (for ESPII) to ensure a greater degree of comparability. When doing this, alpha is calculated as the share of the value of the externalities as a share of the joint economic value of externalities and outputs/inputs. The formula used for alpha is provided in Annex C Table A C.4 and the actual values are provided in Annex C Table A C.5.

Once all indices are computed, annual growth rates in the adjusted metrics are computed and averaged for each country and each index.

Comparing the growth rates of different SPG metrics: Differences become starker when the environment becomes more important

This section focuses on a comparison of the results obtained for each of the three case study countries using six different indices (EATFPI, EATFPO, ESPIO, ESPII, and two versions of WEP (WEPI and WEPO)²⁸). For each index, adjustments are done for six different combinations of environmental indicators, namely adjusting for 1) on-farm GHG emissions only; 2) ammonia emissions from agriculture only; 3) estimated nitrogen run-off and leaching; 4) estimated phosphorus run-off and leaching; 5) a combination of GHG emissions, nitrogen and phosphorus (i.e. exclude ammonia); and 6) a combination of all externalities together. For each specification, two different sets of prices are used. One based on costs from a producer perspective (usually abatement costs derived from the literature) and a proxy of societal costs that goes beyond the costs incurred by producers.

With regards to the construction of the indices, for EATFP metrics shadow prices are used directly to adjust the output/input.²⁹ For all metrics where the weight is set using an exogenous parameter (ESPII, ESPIO, WEPI, WEPO), γ is set as the ratio of the value of the externality(ies) over the joint monetary value of the externalities plus the value of inputs or outputs (i.e. $\gamma = \frac{peE_0}{p_x X_0 + peE_0}$ for ESPII and WEPI and $\gamma = \frac{peE_0}{pyY_0 + peE_0}$ in the case of ESPIO and WEPO). These values of γ are chosen to make these metrics more comparable to the EATFP metrics (Table 1.).

The comparison of growth rates across the different specifications highlights several main aspects across the three case study countries (Table 3-Table 5).³⁰

First, with the exception of New Zealand for the specification focused on GHG emissions when ESPI-O is used, differences between traditional TFP and SPG metrics are small when adjusting for only one environmental dimension individually valued from a producer perspective (columns (1)-(4) of Table 3-Table 5).³¹ This result is intuitive since in these cases the weight of the environmental externality remains relatively small. Therefore, in these cases the majority of the growth in environmentally adjusted productivity metrics is driven by traditional TFP (i.e. changes in output and inputs). This is also consistent with economy-wide estimates of environmentally adjusted TFP, which find that differences with respect to traditional TFP are small when the prices of the externalities imply a small weight of the environment relative to TFP (De Ridder and Lukasz, 2025^[41]).

Second, differences in growth rates across different metrics become larger as more dimensions are included (columns (5)-(6) and (11)-(12)) and as higher environmental valuations are used (i.e. from a societal perspective, columns (7)-(12)). The main reason for this is that higher valuations of individual externalities and additional environmental dimensions all lead to higher weights assigned to the environment. As a result, differences in growth rates between outputs, inputs, and externalities, as well as the properties of indices more apparent.

A third aspect that is common across the different countries is that, generally, growth rates obtained when computing EATFPO exceed all others. EATFPI and ESPII typically provide the second highest values (usually above traditional TFP values). ESPIO, WEPI and WEPO exhibit the lowest growth rates across all countries, and this becomes especially clear when multiple externalities are included.

EATFPO, EATFPI and ESPII require relative decoupling between the externalities and output (EATFPO) or input (EATFPI and ESPII) for their growth rates to exceed that of traditional TFP (the value of TFP for each country is provided in the footnote of the respective Table). Given that in all analysed countries most externalities have grown slower than output and inputs (this is analysed in the next section), the growth rates of these indices exceed that of traditional TFP. The growth rates of EATFPO are higher because in all case study countries output has grown faster than inputs and therefore the differential with respect to the growth rates in externalities is higher.

On the other hand, ESPIO, WEPI and WEPO require absolute decoupling at a specific minimum pace (i.e. that the decrease in the growth rate of externalities is at least as high as the increase in output). Of the three case studies, this has only happened in the Netherlands. This is because output growth was slower (which makes it easier to achieve the condition), there were very large decreases (from a high initial level) in several externalities, and that these were concentrated on externalities with very high societal valuations in the EU (typical ranges for such valuations of ammonia emissions in the EU go from EUR 17 000–EUR 39 500 per tonne (De Vries et al., 2024^[39])). In the cases of Mexico and New Zealand, these rates are lower than traditional TFP because output growth rates were very fast. Despite sometimes-large decreases in the externality intensity metrics in the agricultural sector, the overall level of externalities increased. This leads to a large decrease in the average growth rate when compared with average TFP for these two countries, especially when accounting for multiple externalities and using valuations from a societal perspective.

Table 3. Comparison of average growth rates across different EAP metrics - Mexico

Av. TFP growth = 1.75% SPG measure	(1) GHG - PP	(2) Ammonia - PP	(3) Nitrogen-PP	(4) Phosphorus - PP	(5) GNP -PP	(6) GNPA - PP	(7) GHG - SP	(8) Ammonia - SP	(9) Nitrogen - SP	(10) Phosphorus - SP	(11) GNP - SP	(12) GNPA - SP
EATFPI	1.75	1.75	1.76	1.75	1.75	1.76	1.74	1.76	1.80	1.75	1.79	1.80
EATFPO	1.75	1.80	1.81	1.76	1.84	1.89	2.09	1.85	2.27	1.76	2.93	3.22
ESPII	1.75	1.75	1.76	1.75	1.76	1.76	1.74	1.76	1.80	1.75	1.80	1.81
ESPIO	1.73	1.66	1.64	1.71	1.59	1.52	1.22	1.58	1.20	1.71	0.83	0.75
WEPI	1.74	1.71	1.70	1.73	1.67	1.63	1.47	1.66	1.47	1.73	1.18	1.10
WEPO	1.73	1.68	1.67	1.72	1.63	1.56	1.32	1.61	1.33	1.72	0.88	0.75

Notes: For the analysed time period for Mexico the average annual growth rate of traditional TFP averaged 1.75% p.a. All values represent average annual growth rates (ln growth rates) over the period 1990-2019. GHG refers to on-farm GHG. PP refers to prices for environmental externalities being those from producer perspective. SP denotes prices that consider costs that go beyond abatement at farm-level (e.g. water treatment costs, health costs). GNP denotes the growth rates when GHG, Nitrogen and Phosphorus (i.e. exclude ammonia) are incorporated. GNPA denotes the specification where all externalities are included (GHG, nitrogen, phosphorus and ammonia).

Table 4. Comparison of growth rates across EAP metrics – the Netherlands

Av. TFP growth = 0.38% measure	(1) GHG - PP	(2) Ammonia - PP	(3) Nitrogen-PP	(4) Phosphorus - PP	(5) GNP -PP	(6) GNPA - PP	(7) GHG - SP	(8) Ammonia - SP	(9) Nitrogen - SP	(10) Phosphorus - SP	(11) GNP - SP	(12) GNPA - SP
EATFPI	0.42	0.46	0.42	0.39	0.46	0.53	0.45	1.33	0.44	0.39	0.50	1.36
EATFPO	0.44	0.48	0.43	0.39	0.50	0.61	0.49	3.59	0.46	0.39	0.58	5.26
ESPII	0.36	0.45	0.36	0.32	0.42	0.55	0.39	1.70	0.39	0.32	0.48	1.87
ESPIO	0.34	0.44	0.35	0.32	0.39	0.51	0.36	1.60	0.37	0.32	0.43	1.51
WEPI	0.41	0.50	0.42	0.39	0.46	0.58	0.43	1.59	0.44	0.39	0.50	1.70
WEPO	0.42	0.52	0.42	0.40	0.47	0.60	0.43	1.68	0.45	0.40	0.51	1.81

Notes: For the analysed time period for the Netherlands the average annual growth rate of traditional TFP averaged 0.38% p.a. All values represent average annual growth rates (ln growth rates) over the period 1990-2019. GHG refers to on-farm GHG. PP refers to prices for environmental externalities being those from producer perspective. SP denotes prices that consider costs that go beyond abatement at farm-level (e.g. water treatment costs, health costs). GNP denotes the growth rates when GHG, Nitrogen and Phosphorus (i.e. exclude ammonia) are incorporated. GNPA denotes the specification where all externalities are included (GHG, nitrogen, phosphorus and ammonia). Values of the ESPI specifications (ESPII and ESPIO) need to be compared against a slightly different average growth rate (0.31%).

Table 5. Comparison of growth rates across EAP metrics - New Zealand

Av. TFP growth = 0.21% SPG measure	(1) GHG - PP	(2) Ammonia - PP	(3) Nitrogen- PP	(4) Phosphorus - PP	(5) GNP -PP	(6) GNPA - PP	(7) GHG – SP	(8) Ammonia - SP	(9) Nitrogen - SP	(10) Phosphorus - SP	(11) GNP - SP	(12) GNPA - SP
EATFPI	0.29	0.20	0.23	0.21	0.31	0.31	0.36	0.15	0.23	0.21	0.38	0.31
EATFPO	0.35	0.20	0.24	0.22	0.39	0.39	0.50	0.16	0.24	0.22	0.55	0.55
ESPII	0.22	0.11	0.14	0.12	0.25	0.24	0.30	0.07	0.14	0.12	0.33	0.28
ESPIO	-0.10	0.06	0.07	0.10	-0.15	-0.19	-0.25	-0.28	0.07	0.10	-0.29	-0.58
WEPI	0.16	0.18	0.20	0.21	0.14	0.11	0.12	0.00	0.20	0.21	0.10	-0.11
WEPO	0.15	0.18	0.20	0.21	0.13	0.10	0.10	-0.03	0.20	0.20	0.09	-0.15

Notes: For the analysed time period (1990-2021) the New Zealand average annual growth rate of traditional TFP averaged 0.21% p.a. GHG refers to on-farm GHG. PP refers to prices for environmental externalities being those from producer perspective. SP denotes prices that consider costs that go beyond abatement at farm-level (e.g. water treatment costs, health costs). GNP denotes the growth rates when GHG, Nitrogen and Phosphorus (i.e. exclude ammonia) are incorporated. GNPA denotes the specification where all externalities are included (GHG, nitrogen, phosphorus and ammonia). Values of the ESPI specifications (ESPII and ESPIO) need to be compared against a slightly different average growth rate (0.11%).

Growth rates of different SPG metrics and their drivers and illustration using one specification

This subsection looks at the evolution over time of the different components of the SPG metrics (output, input and negative externalities) qualitatively describing some of the drivers of the observed trends in three case study countries (Mexico, The Netherlands, and New Zealand). These drivers are then used to explain observed trends in one of the environmentally adjusted metrics over time (EATFPI) focusing with a combination of all externalities using societal valuations. Finally, a simple decomposition of the metric is provided (see Annex C for a description of the steps) using the assumed environmental values used in the valuation from a societal perspective as an illustration of how the interpretation of SPG metrics could be simplified for non-specialist audiences.

The choice of EATFPI using all externalities and societal perspective for the valuation is not an endorsement of this specific approach nor of the use of a societal perspective to value the externalities. The EATFPI was chosen as the decomposition is intuitive and valuations from a societal perspective were chosen such that changes are graphically more visible. As seen in previous sections, differences between TFP and SPG are much smaller when using producer-centric valuations.

Mexico case study

Narrative of trends in output and agri-environmental indicators (AEIs)

Between 1990 and 2022, the gross value of Mexican agricultural output more than doubled. Increases in the use of inputs were necessary to allow for this expansion, but overall outputs grew at a much faster rate than inputs. This resulted in Mexico exhibiting an average TFP growth rate of 1.75%, substantially above of the OECD average of approximately 1%. Doubling production, however, led to an increase in negative environmental externalities, though these grew at a much slower rate than output increases. However, despite the fast increase in TFP during the observed period, recent analyses find substantial spatial heterogeneity in TFP growth and that low technical efficiency has constrained further productivity growth (Ortega Diaz and Guerrero, 2025^[42]).

Reforms starting in the 1980s and continuing through the 1990s led to profound changes in the Mexican agricultural sector. Prior to the reforms that started in the 1980s, Mexican agriculture was heavily protected, characterised by a policy mix that consisted of Agricultural tariffs and quotas, minimum prices for key staple commodities, input subsidies and a more extensive role of state-owned enterprises (OECD, 2024^[43]). Liberalisation of the sector started in the 1980s which led to large decreases in price support (*ibid*). The signature of the North American Free Trade Agreement (NAFTA) in 1994 led to a phasing out of state intervention in two of the main staples (maize and beans) (Anderson and Valdés, 2007^[44]; OECD, 2024^[43]) as well as a major shift in the producer support system, away from price support and towards direct payments (e.g. *Programa de Apoyos Directos al Campo* - PROCAMPO). The very deep trade reforms pursued by Mexico during this time created an incentive for a higher degree of export orientation in Mexican agriculture and the reduction in input subsidies increased incentives to use inputs more efficiently. Over time this was a key driver of output growth.

In 1995, however, Mexico experienced its worst financial crisis since the 1930s and a substantial devaluation of its national currency, causing economic hardship to the overall population, but also contributing to an improvement of its sectoral trade balance (Ros, 2001^[45]; Mckenzie World Bank, 2006^[46]; Puyana, 2012^[47]). In addition to these factors, technology (e.g. improved varieties) also played an important role for specific commodities (e.g. wheat) (Ortiz et al., 2008^[48]) and a structural shift towards higher value-added commodities (e.g. fruits, vegetables, dairy, poultry³²) also contributed to the large increases in output.

The performance evolution of agri-environmental indicators in the 1990s was mixed. Shifts away from input subsidies, combined with the devaluation of the national currency following the 1995 crisis initially put a downward pressure on input use (especially fertiliser use),³³ which led to lower levels of nutrient surplus for most of the 1990s. However, these started to rebound towards the end of the 1990s, with foreign demand and the recovery from the crisis playing a role. GHG emissions also remained relatively stable between 1990 and 2000, mostly as a result of decreases in input use and relatively stable intensity of livestock, with livestock units per hectare remaining constant during the 1990s (FAO, 2025^[49]). Nevertheless, while overall livestock units remained constant over the period, there were large shifts in composition, with dairy herd sizes increasing by 24% and poultry units increasing by almost 60%, while non-dairy cattle herds decreased by 11%, caprine herds decreased by 16.3%, and horses decreased by 33% (Gobierno de México, 2025^[50]).

From 2000 to 2022, output levels continued to grow and many of the trends observed in the 1990s continued. There was a continued shift towards high-value-added products (dairy, poultry, fruits and vegetables) and an increase in the adoption of certain technologies, such as irrigation technologies. According to Ortega Díaz and Guerrero (2025^[42]), the adoption of such technologies is associated with large productivity impacts, with a 1% increase in traditional and drip irrigation systems increasing TFP growth by 3.8% and 5%, respectively. These factors led to a sustained increase in output levels. However, there were also several shocks that explain some of the dips observed in Figure 1 panel (a). Specifically, there were a succession of shocks in the aftermath of the global financial crisis, including the 2011-12 drought (the worst in 70 years) (Ortega-Gaucin, Bartolón and Bahena, 2018^[51]; Murray-Tortarolo and Jaramillo, 2019^[52]), the avian influenza outbreak of 2012 (Acosta, Barrantes and Ihle, 2020^[53]), the porcine epidemic diarrhoea virus (PED) outbreak of 2013-2014, and the aggressive coffee leaf rust outbreak that started in 2012 (Torres Castillo et al., 2020^[54]).

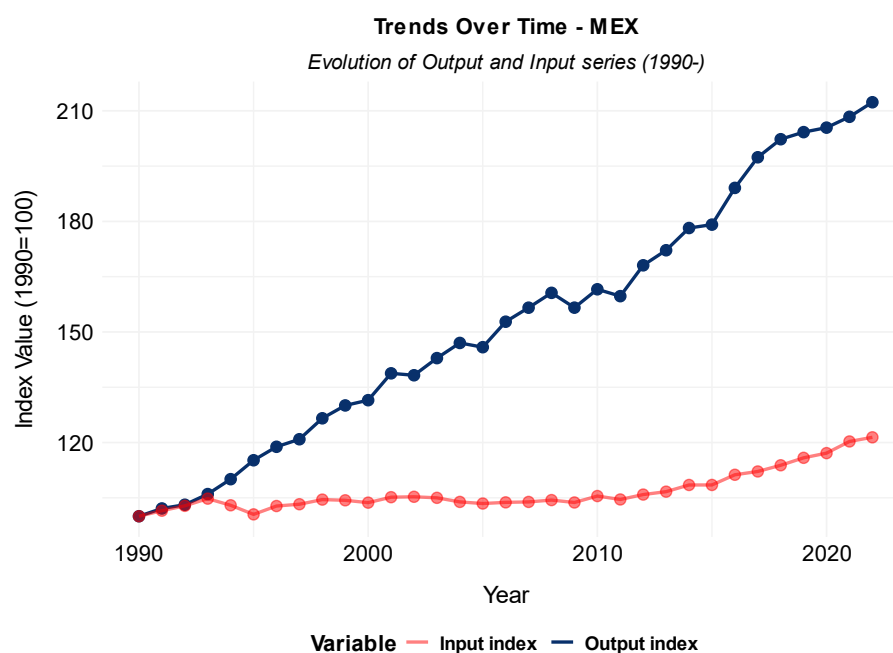
From 2000 to 2022, agri-environmental indicators started to change more significantly. GHG and ammonia emissions were driven mostly by structural factors and herd sizes. Between 2000 and 2015, there were large increases in dairy cattle (18%), non-dairy cattle (9%), sheep (45%) and swine (6%) herds as well as large increases in poultry (45%) (Gobierno de México, 2025^[50]). With regards to nutrient inputs, two main sub-periods (between 2000 and 2009; and 2009-2022) can be identified, as shown in Figure 1 (panel c) and

Figure 2 (panel b). Between 2000 and the global financial crisis, there was a substantial increase in international fertiliser prices, with urea prices increasing fivefold between 1999 and 2007 and DAP prices more than doubling (World Bank, 2025^[55]). Increasing fertiliser prices combined with a depreciation of the Mexican peso over this period put downward pressure on fertiliser use in the absence of subsidies. This trend was particularly noticeable for Urea.

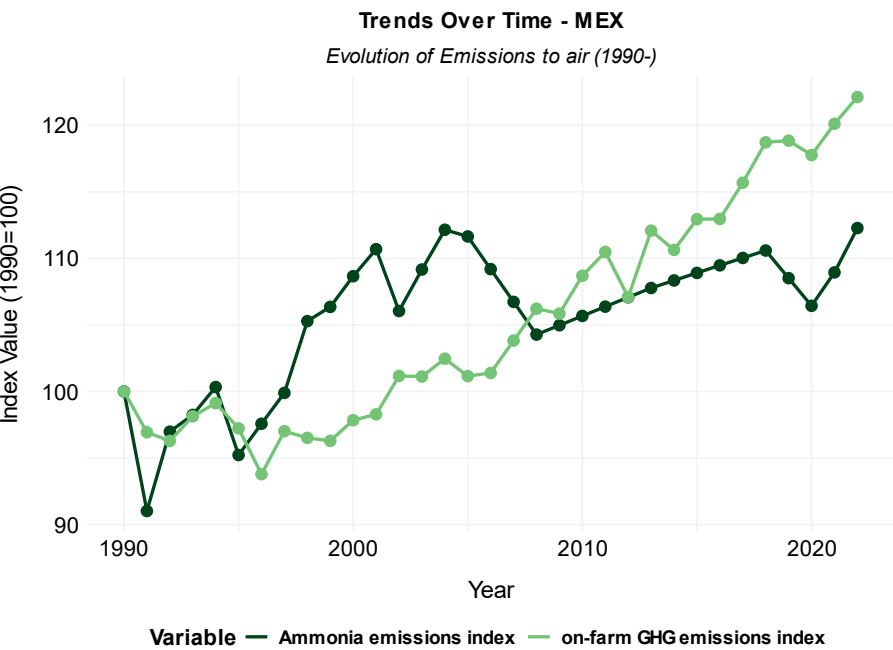
The smaller increase in Di-ammonium Phosphate (DAP) prices led to increases in the use of phosphorus fertilisers. In 2008, following the global financial crisis and phosphate rock price spike, both phosphorus and nitrogen inputs (the latter to a lesser extent) decreased. Since 2009, declining international prices (especially for Urea) and a stable exchange rate (2010-2014) led to an increase in urea use, with this increase continuing until 2021, before a dip being observed as a result of Russia's war of aggression against Ukraine. Since 2018, a fertiliser programme aimed at smallholders was introduced (OECD, 2024^[43]) and could add pressure to increase nutrient inputs, but its effects on nutrient inputs are not yet clearly visible (Figure 1, panel c).

Figure 1. Trends in output, input and externalities (Mexico - 1990-2019)

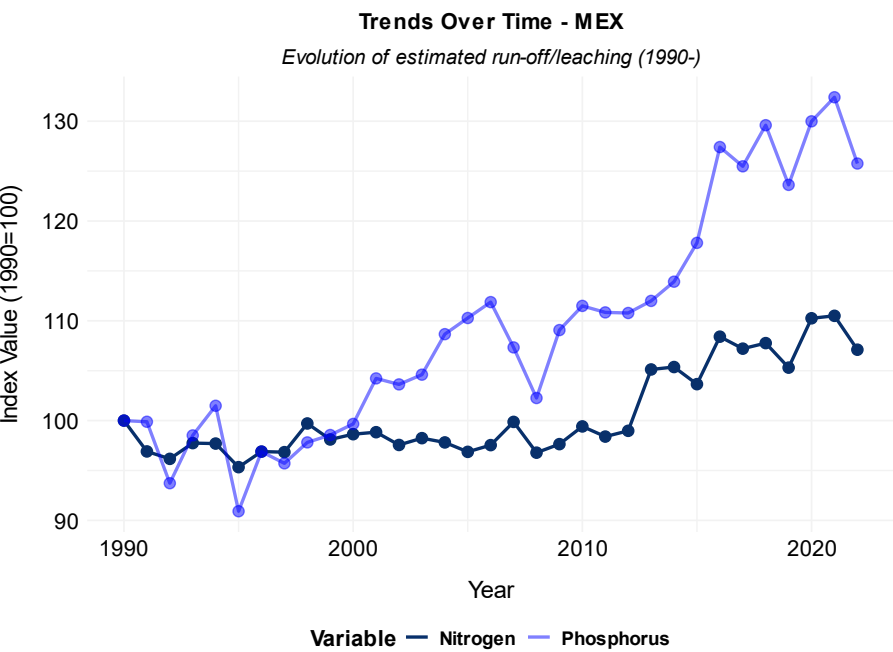
a. Output and inputs



b. Emission of greenhouse gases and ammonia



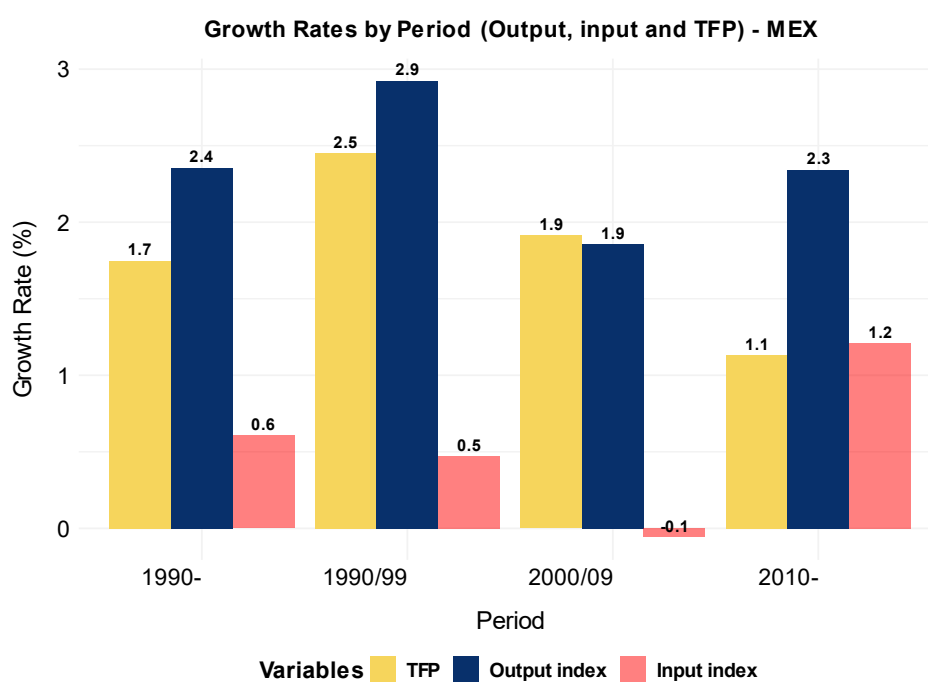
c. Estimated nutrient run-off and leaching



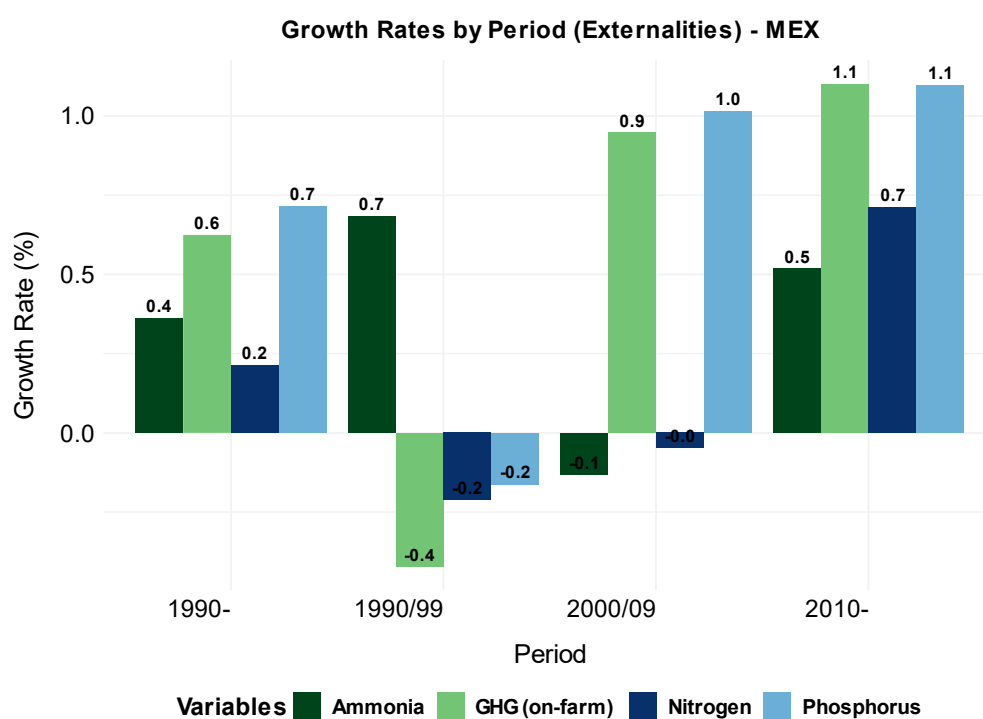
Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

Figure 2. Average growth rates by decade in Mexico

a. Average TFP, output and input growth (by subperiod)



b. Average growth of externalities (by subperiod)



Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

EAP trends over time and decomposition

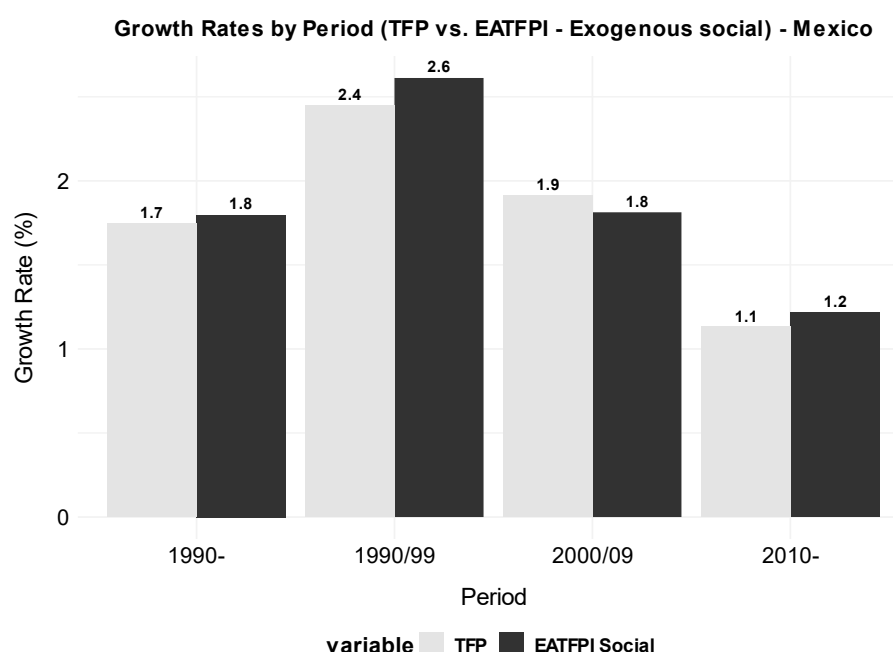
Mexico has had one of the highest TFP growth rates across OECD countries, with an average TFP growth rate of 1.75% p.a. Using the EATFPI specification, even when using valuations of the externalities from a societal perspective, it does not change the results very much with the differential between the two (TFP and EATFPI) metrics never exceeding 0.2 p.p. across all the subperiods (see Figure 3, panel a).

However, the results vary by decade and the importance of environmental externalities in the SPG metric have increased over time (Figure 3, panel b). In the 1990s, the downward pressures on input use (especially fertiliser) discussed above and the fact that livestock intensity remained broadly constant meant that the output expansion was able to occur alongside moderate increases in the externalities. As a result, in the 1990s the environmentally adjusted growth rates exceeded traditional TFP. This started changing in the 2000s when output expansion was also accompanied by herd size expansion and an increase in the intensity of fertiliser use.

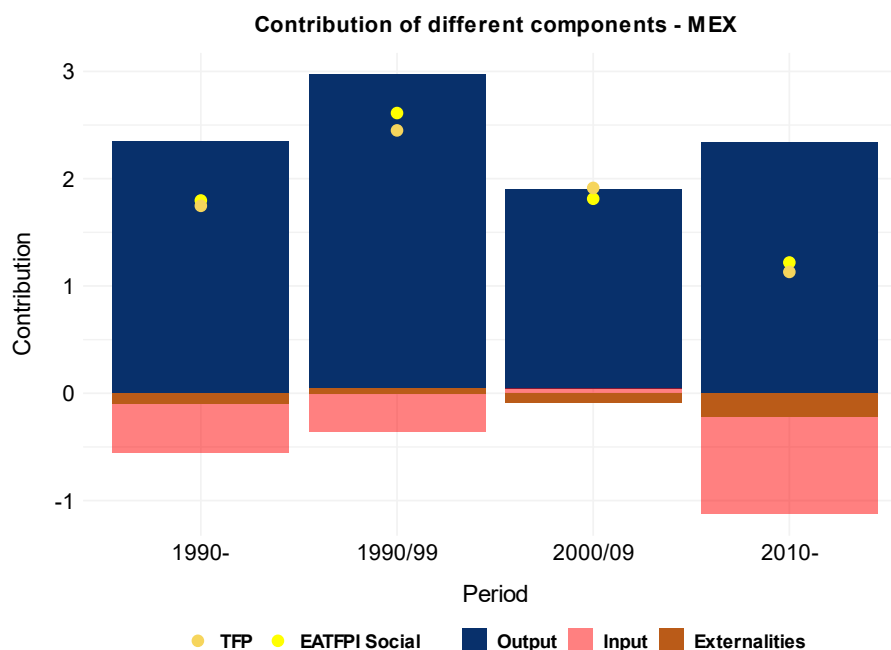
Nevertheless, the negative contribution of environmental externalities remains small overall when compared to the contributions of output and input (Figure 3, panel c). Increases in GHG emissions was the externality associated with the highest negative (GHG) contribution to the SPG metric.

Figure 3. TFP vs. EATFPI (Social) in Mexico (by subperiod)

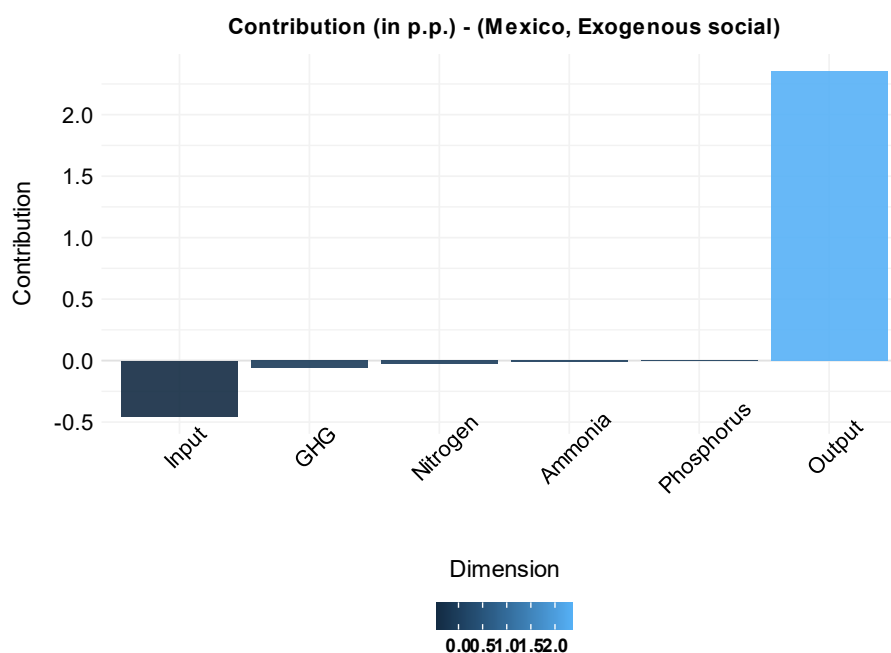
a. Average TFP and EATFPI growth rate (by subperiod)



b. Contribution to average EATFPI growth rate (by subperiod)



c. Contribution of individual externalities to average EATFPI (in p. p.)

**The Netherlands case study***Narrative of trends in output and AEIs*

During the analysed time period (1990-2022), several important developments in the Dutch agricultural sector enabled increases in output with limited increases in input use. At the same time, large reductions in all the considered externalities (GHG emissions, nutrient balances, and ammonia emissions) were

achieved, especially during the 1990-2010 period (Figure 4). However, despite reduction in the absolute levels achieved since the 1990s, the absolute values of some of these externalities (e.g. nitrogen balances) remain substantially above OECD averages and have led to the urgent need to reduce nitrogen deposition via ammonia emissions after the 2019 Court ruling (OECD, 2023^[56]).

During the 1990s, output growth was positive but volatile, with most of traditional TFP growth explained by the decreases in input use (Figure 4, panel a; Figure 5 panel a). During the same period, a steady decrease in the GHG emissions and nutrient inputs was observed, alongside a rapid decrease in agricultural ammonia emissions which approximately halved during the decade (Figure 4 panels b and c) and Figure 5 panel b).

Several factors explain these trends. At the EU level, the 1990s marked the beginning of the “MacSharry” CAP reform which reduced the levels of price support and led to a shift in income support (OECD, 2024^[43]). It also marked the introduction of the Nitrates Directive, with a renewed focus on some agri-environmental externalities (OECD, 2015^[57]). At the same time, technological improvements (e.g. improved crop genetics) contributed to yield gains (Rijk, van Ittersum and Withagen, 2013^[58]) alongside structural changes under way in the Dutch agrifood sector, such as a shift towards export-oriented agriculture, horticulture, alongside farm consolidation (with a 20% reduction in the number of holdings during the 1990s; (FAO, 2025^[59])) enabled output gains alongside reductions in inputs.

On the environmental side, several factors contributed to the reductions in the externalities. Structural shifts in the livestock sector contributed to reducing environmental pressures, with large decreases witnessed in herd sizes of cattle (17%) and increases in poultry (12%) between 1990 and 2000 (van der Net et al., 2024^[60]). EU level policies, such as the 1992 “MacSharry” CAP reform and the Nitrates Directive, led to reductions in price support and to a shift towards direct payments (still tied to land and animals), but also increased incentives for greater efficiency in the use of resources and inputs. National policies also played an instrumental role, especially for ammonia and nutrient balances. In 1993, legislation was introduced to enable a system of environmental permits. The 1990s also saw a substantial tightening of manure policy. The soil protection Act of 1986 authorised restrictions on the quantity of applied manure (Backus, 2017^[61]). In addition, more stringent regulations related to application methods, timing and storage of manure and mineral fertilisers were put in place between 1993 and 1999 (Fraters et al., 2004^[62]). This included a requirement to ensure that manure storage infrastructure was covered (a subsidy was introduced to support this), and a requirement to inject manure directly (Fraters et al., 2004^[62]; Backus, 2017^[61]). In 1998, the Mineral Accounting system (MINAS) was introduced and it included a penalty for farmers exceeding a certain threshold of nutrient surplus (Backus, 2017^[61]). These factors contributed to incentivise the adoption of improved manure storage and increased adoption of specialised fertiliser spreading equipment. Combined with favourable structural trends, these contributed to a large reduction in ammonia emissions, as well as phosphorus and nitrogen surpluses.

In the 2000s, output trends were severely affected by several large shocks and policy changes which led to a negative average growth rate of output during the period. Between 2001 and 2003, a large outbreak of avian influenza which led to the culling of 30 million birds (Stegeman et al., 2004^[63]) and a foot-and-mouth disease outbreak (Stegeman et al., 2004^[63]) all contributed to decreased levels of output in the first half of the 2000s, before recovering in the second half. This was followed by a period of strong Euro appreciation between 2002-2007 followed by the global financial crisis. On the policy front, the 2003 CAP reform also changed the way in which support was provided, increasingly shifting to direct payments and introducing cross-compliance. At the structural level, the trends of increased export orientation, specialisation and farm consolidation continued during this period.

Agri-environmental indicators continued to improve substantially (compared to 1990 levels) during the 2000s. At the structural level, livestock numbers continued to decrease, especially for pigs (van der Net et al., 2024^[60]). With regards to policies, the 2003 CAP reform decoupled most support from production and made cross-compliance mandatory (OECD, 2024^[43]). Domestically, manure and ammonia regulations

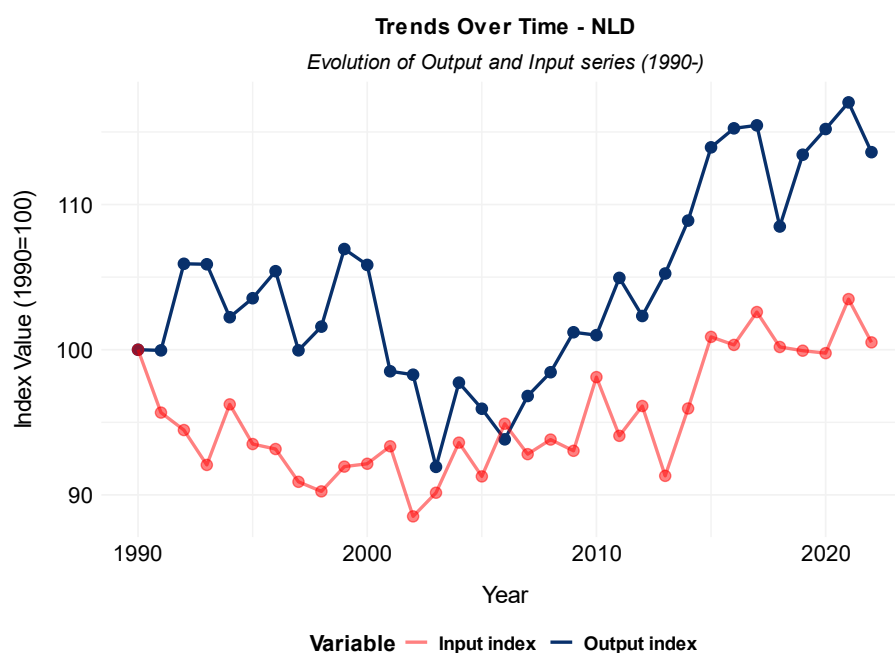
continued to evolve and were key drivers behind improved nutrient balances and further ammonia reductions. The MINAS system continued until 2005, after which it was abandoned following an European Court of Justice (ECJ) ruling in 2003, which led the Netherlands to switch to a regulation based on application rates, where the ceilings were gradually reduced over time (Backus, 2017^[61]). This was complemented with a system of production rights for pigs and poultry introduced in 2006 (RIVM, 2024^[64]) aimed at capping growth in livestock and fostering efficiency gains. In addition to this, the fertiliser price hike of 2008/2009 also put a downward pressure on nutrient surpluses.

In the most recent period analysed (2010-2022),³⁴ output expanded significantly compared to the previous decade. The continued trend of specialisation, shift to high-value added sectors, together with the designation of the agrifood sector as a “top sector” for innovation in 2011 and the roll-out of technologies all contributed to this trend. While data is not nationally representative, among a sample of Dutch arable farmers the adoption rates of specific precision technologies such as sensors and variable rate application technologies all increased very fast since 2010 (Kempenaar, 2020^[65]). In 2018, however, the Dutch agricultural sector faced a severe drought, which explains the dip in production (Prins et al., 2018^[66]).

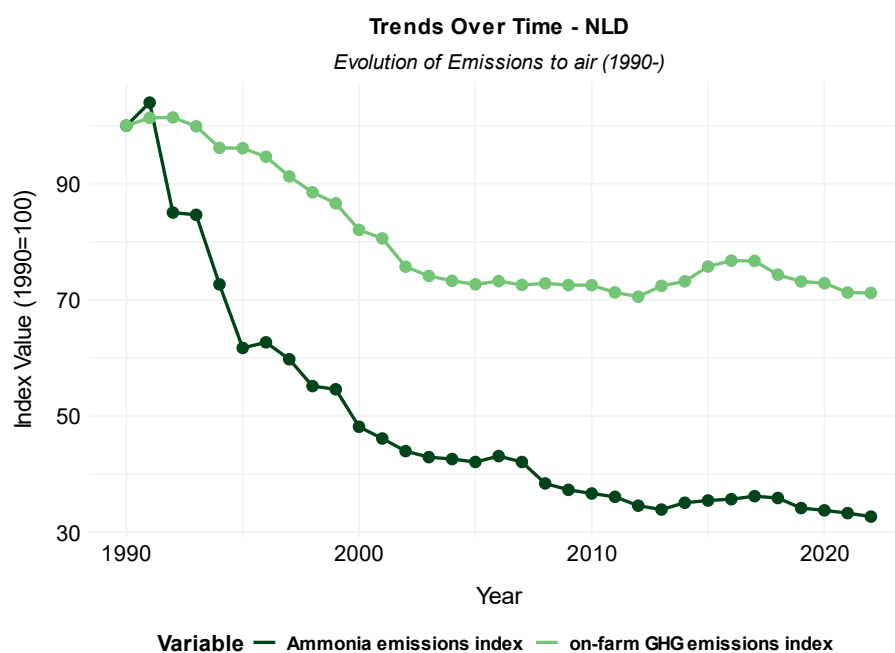
Progress on agri-environmental indicators has slowed down since 2010 and improvements have been less consistent. Previous efficiency means that further reductions are more costly and approaching technical limits. However, different factors have also led to both positive and negative incentives to improve agri-environmental performance. At the EU level, the CAP greening was introduced in the 2013 CAP reform which tied a share of direct payments to greening measures and introduced additional elements, such as eco-schemes. Beyond changes in the CAP, the increased adoption precision agriculture (sensors, variable rate application), transition towards improved housing in the poultry sector, and the introduction of the manure removal system in 2014 requiring farmers to process part of their manure and the increase in manure exports³⁵ also put a downward pressure on ammonia emissions and nutrient inputs (van der Net et al., 2024^[60]; Backus, 2017^[61]; Kempenaar, 2020^[65]; RIVM, 2024^[64]). Nevertheless, the abolition of the EU milk quotas in 2015, led to an increase in the number of dairy cattle reversing the decreasing trend that had been observed in the previous decades (RIVM, 2024^[64]; Berkhout, Meulen and Ramaekers, 2023^[67]). In order to counter this trend, following a compulsory reduction in livestock in 2017, the phosphate production rights has been applied to the dairy sector since 2018 (RIVM, 2024^[64]). In 2015, the Government also introduced the Programmatic approach to Nitrogen (PAS) which allowed permits based on anticipated future reductions. In 2019, a court ruling found it to be invalid casting doubt on the whole system. Finally, in 2022, a decrease in the level of phosphorus inputs is observed, probably driven by the increases in international prices observed following the beginning of Russia’s war of aggression against Ukraine.³⁶

Figure 4. Trends in output, input and externalities (the Netherlands - 1990-2019)

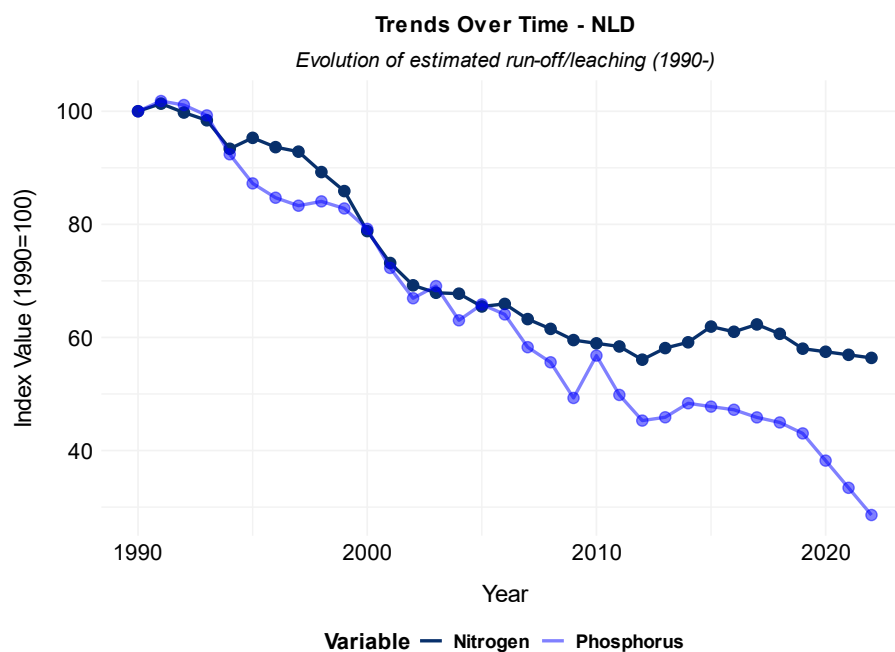
a. Output and inputs



b. GHG and ammonia



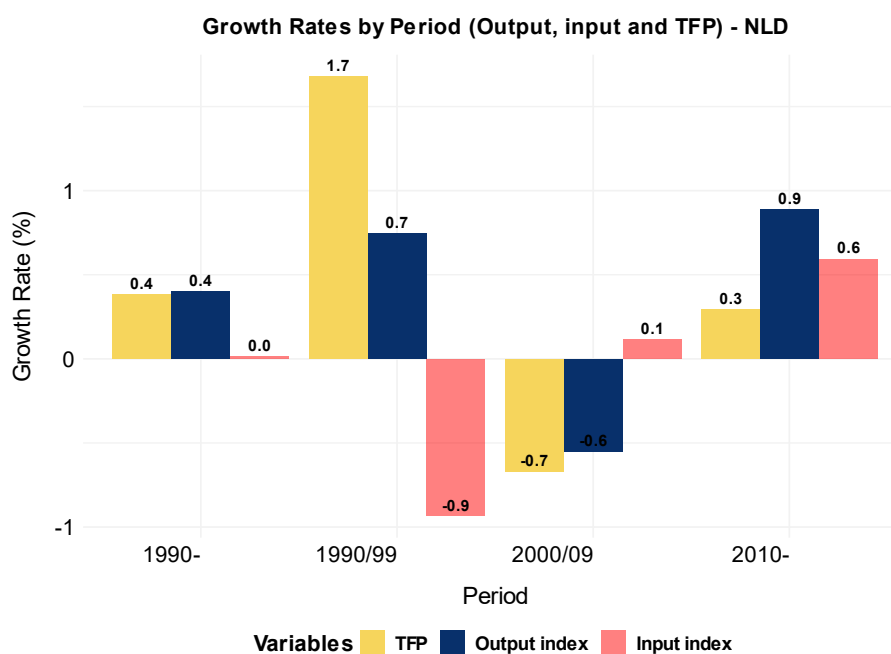
c. Estimated nutrient run-off and leaching



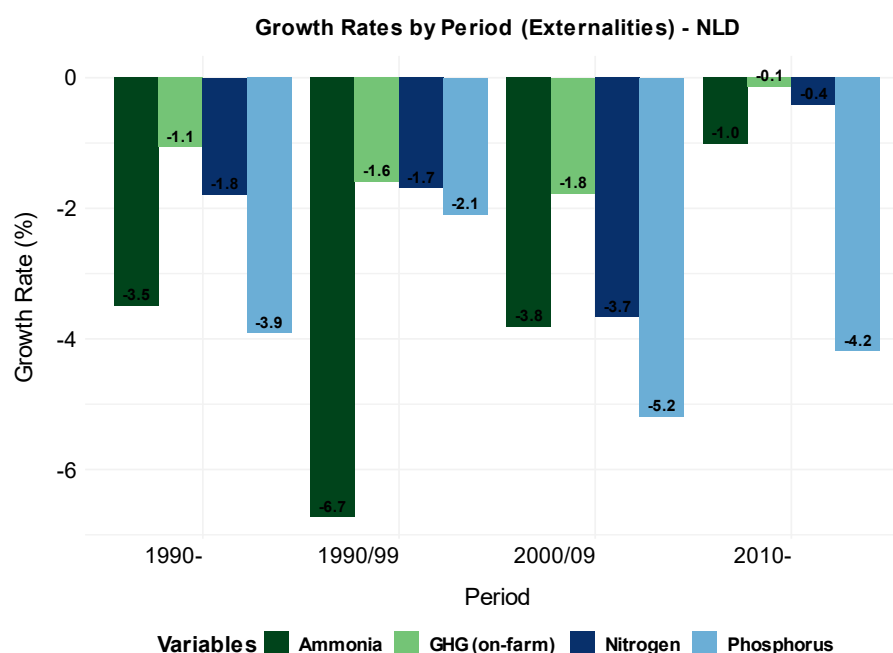
Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

Figure 5. Average growth rates by decade

a. Average TFP, output and input growth (by subperiod)



b. Average growth of externalities (by subperiod)



Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

SPG trends over time and decomposition

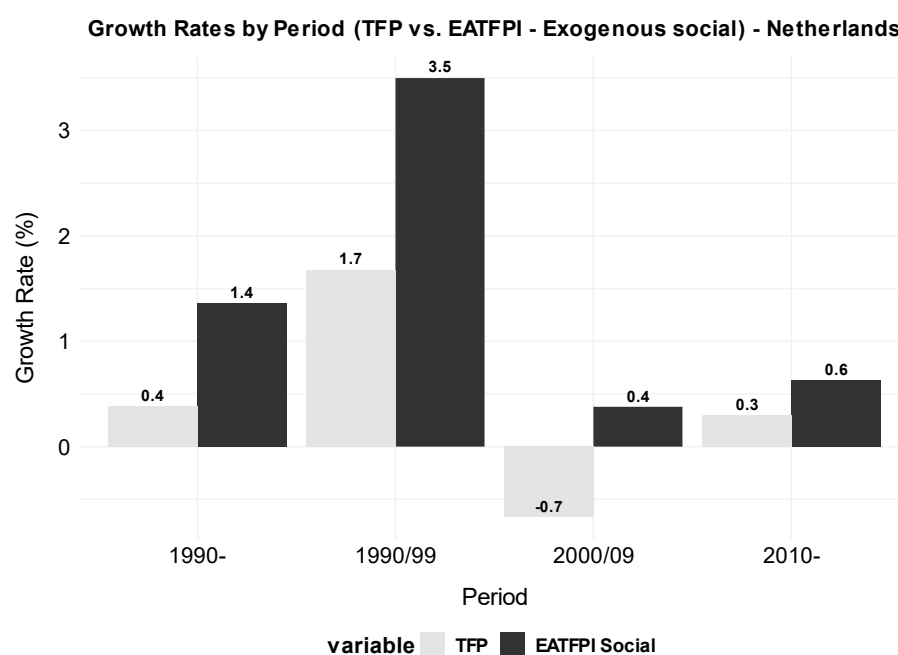
In the case of the Netherlands, moving from TFP metrics to SPG leads to very large differences in the growth rate. TFP growth averaged a 0.38% p.a. increase between 1990-2022. This increases to 1.4% p.a. when using societal valuations (see Figure 6, panel a). This is because, while the absolute level of externalities remains high in some dimensions (e.g. nitrogen inputs and balances), their current levels are below their levels in the 1990s.

However, differential between SPG and TFP growth rates differ substantially across sub-periods and declines over time, with the biggest differences being observed in the 1990-2010 period. This coincides with the period where many regulations (especially on manure, fertiliser and ammonia) were introduced, which led to both structural changes and adoption of several practices which contributed to reducing the externalities. As an example, while the TFP-EATFPI differential was 1.8 p.p. in 1990s, this decreased to 0.3 since 2010. During the 1990s, when there were noticeable improvements across all the considered externalities (especially ammonia which halved over a 10-year period). This led to an estimated EATFPI growth rate (using social valuation of ammonia) that is double the TFP growth rate. A similar trend can be observed in the 2000s. However, since 2010, improvements in agri-environmental performance have slowed (with the exception of phosphorus inputs) substantially and the difference between the two metrics is substantially smaller and their contribution to the overall EATFPI is declining over time (Figure 6, panels a and b).

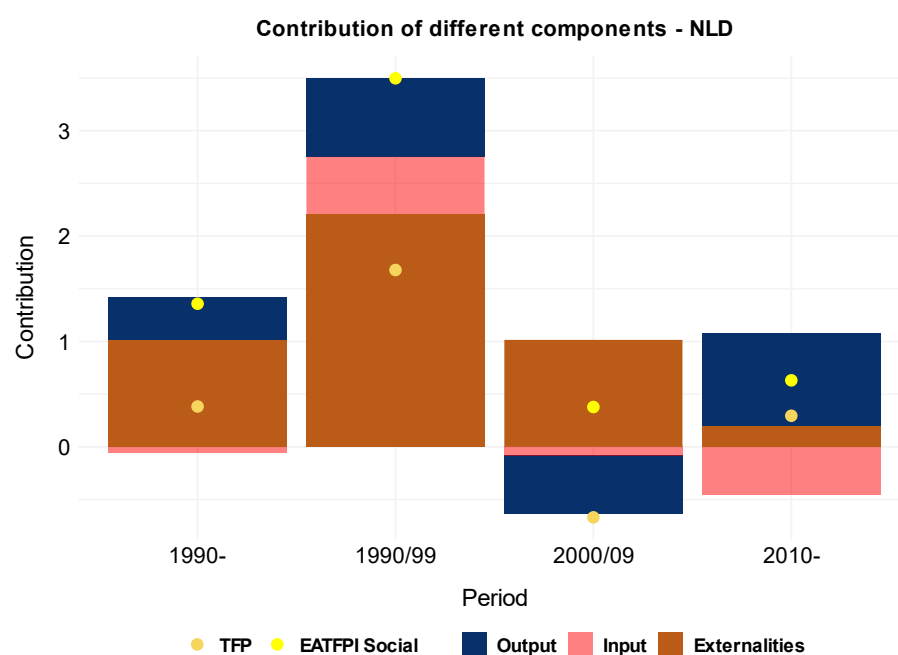
Importantly, the contribution of the different externalities to the adjustment is uneven, with a very large proportion of the improvements driven by reductions in ammonia emissions (Figure 6, panel c). Improvements in estimated Nitrogen run-off and leaching and GHG emissions also contributed somewhat to improving the EATFPI measure. The fact that the Netherlands substantially reduced its ammonia emissions (which have, a very high social valuation in the European context) is one of the key reasons explaining the very positive adjustment of the EATFPI.

Figure 6. TFP vs. EATFPI (Social) in the Netherlands (by subperiod)

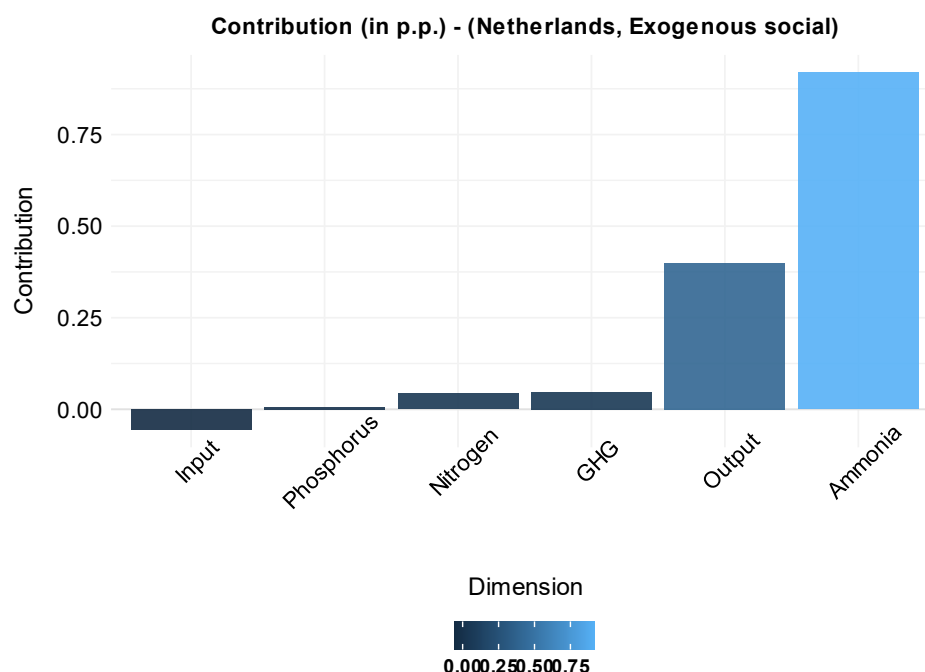
a. Average TFP and EATFPI growth rate (by subperiod)



b. Contribution to average EATFPI growth rate (by subperiod)



c. Contribution of individual externalities to average EATFPI (in p. p.)



Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

New Zealand

Narrative of trends in output and AElS

There was a substantial expansion of output in New Zealand between 1990 and 2022, with value of output (measured in constant international USD) growing at an average annual growth rate of 1.6% (Figure 7). The expansion of output in New Zealand was driven primarily by growth in the dairy and horticultural sectors³⁷ which supplied international markets, with the proportion of milk produced being exported exceeding 80%.³⁸ Such an expansion in output largely fueled by international demand for dairy products led to fast rate of input growth (at 1.4% a year) and was accompanied by an increase in all the externalities included in the analysis, namely agricultural GHG emissions and ammonia emissions, as well as estimates of nitrogen and phosphorus run-off and leaching. However, the trends in these indicators were very uneven across indicators during the analysed period.

A wave of profound economic reforms started in 1984, and these led to profound changes in New Zealand's economic and agricultural policy. These reforms dismantled a system of agricultural support that was based on widespread use of subsidies, which heavily benefitted sheep and wool, input subsidies and import controls (Hugonnet, 2025^[68]; Ge, 2025^[69]). The reforms in the agricultural sector were accompanied by economy-wide reforms that resulted in a devaluation of the exchange rate, a liberalisation of the financial sector, and a reduction in restrictions affecting capital flows (Hugonnet, 2025^[68]).

These reforms had two effects that explain observed trends. First, they fully exposed New Zealand's agricultural sector to market forces and set the basis for an export-based agricultural sector. Second, it fundamentally changed the incentive structure of the agricultural sector. This particularly affected sheep producers, who were the most reliant on Government support. The changes in trade policy and increased exposure to international markets also created incentives for the intensification of agriculture.

The reforms of the mid-1980s led to a fast transition away from sheep and wool towards dairy throughout the 1990s, the period where New Zealand achieved the highest traditional TFP growth rates (1.3% p.a.), mostly driven by output growth (1.7% p.a.). Much of this growth can be attributed to the structural changes in the agricultural sector. During the 1990s, sheep populations decreased by approximately 20% while those of dairy cattle increased by approximately 25% (Ministry for the Environment, 2025^[70]; StatsNZ, 2024^[71]). The 1990s also saw the beginning of the adoption of a more intensive pasture model, substantial genetic improvements in perennial ryegrass, as well as a change in the selection criteria for dairy cattle genetics (Breeding worth index) in the mid-1990s that put a greater emphasis on profitability (Chapman et al., 2025^[72]; Harris, 2005^[73]). All of these factors contributed to the high TFP growth rate observed during this period, which would have probably been higher if not for negative shocks during 1997-1999, when New Zealand was hit by two consecutive droughts with large economic impact (Salinger and Porteous, 2014^[74]) and the aftermath of the Asian financial crisis.

These structural shifts put additional pressure on the environment. The shift from sheep to dairy and a more intensive model of agriculture led to an increase in ammonia emissions and GHG emissions, as well as higher nitrogen and phosphorus inputs. GHG emissions, in particular, were closely aligned with herd sizes. In an attempt to mitigate the environmental consequences of economic activity, the Government introduced the Resources Management Act in 1991 which replaced and strengthened a number of existing statutes related to environmental management and planning. The Government also operated two novel soil quality monitoring programmes starting in the mid-1990s which tested different indicators and identified soil and land use management combinations of risk (Sparling et al., 2004^[75]; Sparling and Schipper, 2004^[76]).

In the 2000s, rates of overall TFP growth in New Zealand decreased substantially, becoming negative. While output growth increased (to 2.2% p.a.), so did input growth (2.4% p.a.). The overall trend of decreasing sheep populations and increasing dairy populations continued in the 2000s, a further decline of approximately 30% in sheep population and a 36% increase in dairy cattle herds. The overall trend of intensification of agriculture also continued. In addition to the structural trends, the 2000s also saw the creation of Fonterra, the country's largest cooperative), as well as the signature of two important Free Trade Agreements, with China (2008) and ASEAN (2010). These changes opened new markets, contributed to a better positioning of New Zealand in global value chains (especially dairy) and accelerated export growth, positively contributing to output trends (Verevis and Üngör, 2021^[77]). With regards to negative pressures, the financial crisis in 2008/09 financial crisis put a downward pressure on output and also led to reductions in inputs, partially as a result of higher fertiliser prices.

During the 2000-2009 period, environmental externalities grew (with the exception of phosphorus), but their growth rate decreased. While ammonia emissions continued to grow in a similar fashion to herd sizes, GHG emissions reached a peak around 2005 (as a result of substantial gains in emission intensity of GHG emissions of approx. 20% between 2000-2009). Improved emission intensities were a result of several factors, including continued intensification of cattle production, improved genetics, improvements in animal health. During the 2000s, nutrient balances also increased at a much lower rate and three factors help explain observed trends in the estimated nutrient run-off and leaching. First, the more extensive roll-out of nutrient management tools (Ministry for the Environment and Ministry for Primary Industries, 2021^[78]) and soil testing (Flett et al., 2004^[79]) were important factors to increase adoption of variable rate fertiliser application. Second, the largest dairy cooperative committed its farmers to the Dairying and Clean streams accord in 2003, which effectively required farmers to adopt several improved practices by 2012, including nutrient and effluent management (Quinn et al., 2009^[80]). Finally, shocks also played an important role as in 2008 fertiliser markets were simultaneously hit by the financial crisis, surges in grain demand, high energy prices, and supply restrictions of phosphate rock. Together these led to a very rapid increase in fertiliser prices (IFDC, 2008^[81]) which led to a short-term reduction in fertiliser use.

Since 2010, overall TFP growth has been negative with output continuing to grow at a high pace (1.1% p.a.), but not faster than input growth (1.4% p.a.). Since 2010, sheep population continued to decline and

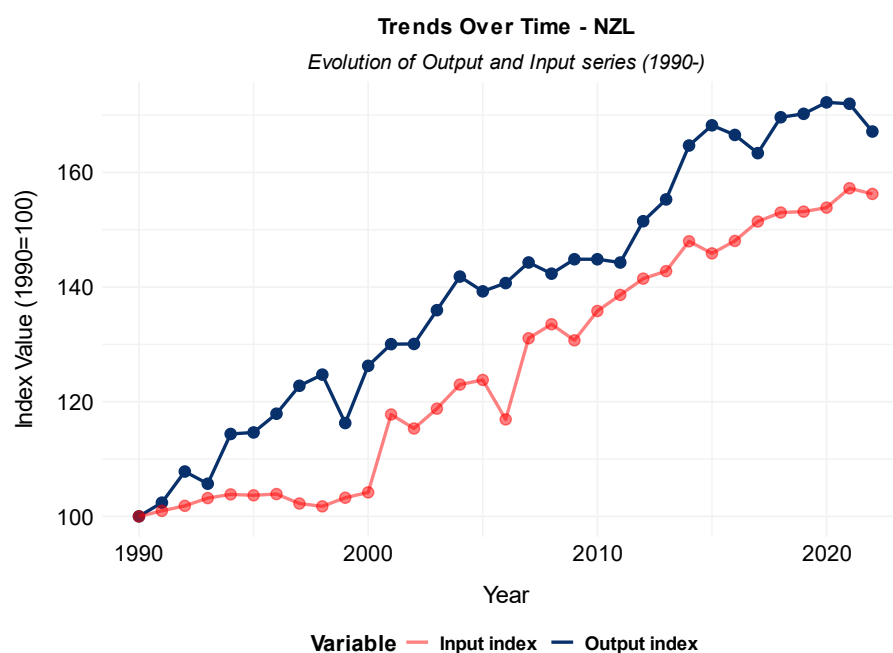
dairy herd sizes increased until they peaked *circa* 2014, before starting to gradually drop. The trend of input intensification continued, but gains to pasture productivity also started to markedly slow since 2011 (Chapman et al., 2025^[72]). The COVID pandemic in 2020 also put a downward pressure on output growth towards the end of the analysed period.

In the 2010-2022 period, the overall growth in ammonia emissions slowed down substantially. Following an increase until 2014, ammonia emissions seem to have stabilised. With regards to GHG emissions, despite increasing herd sizes until 2014, improvements in emission intensity prevented overall levels of GHG emissions to go beyond levels observed in 2005. This period saw increases of adoption of agricultural technologies as well as the introduction of new policies. The adoption rates of urease inhibitors increased from 6.9% to 60% between 2010 and 2022 (Ministry for the Environment, 2025^[70]) and studies (not nationally representative) have highlighted an increase in the uptake in digital technologies, with the likelihood of adopting digital technologies increasing by approximately 300% between 2008 and 2018 in some samples of dairy farmers (Yang et al., 2021^[82]). Another important aspect was the National Policy Statement for Freshwater Management (NPS-FM). The NPS-FM was first introduced in 2011³⁹ and from 2011 onwards, regional authorities introduced policies under the NPS-FM that more stringently managed discharges from intensive agriculture. Beyond these factors, a decrease in GHG emissions was also observed during COVID-19.

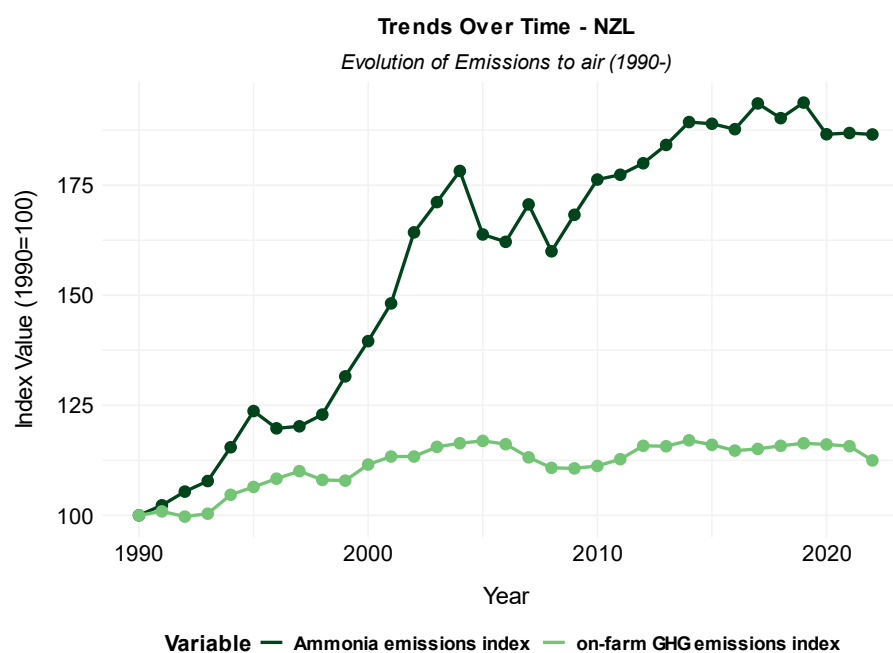
Very different trends were observed for nutrients, as nitrogen inputs started to decrease post-2010 and phosphorus inputs have been more volatile, hovering around their 2010-2011 levels. As highlighted by Macintosh, McDowell and Thiange (2025^[83]), the 2010s were marked by an improvement in the adoption of good management practices in dairy sector partly as a result of the dairy and Streams Accord and the subsequent Sustainable Dairying water accord of 2013. These agreements placed additional emphasis on riparian management plans and water use management. There was also the continued adoption of technologies such as urease inhibitors. In addition, policies and shocks, including the global COVID-19 pandemic, higher fertiliser prices following the start of Russia's war of aggression against Ukraine, and the announcement and introduction of the nitrogen cap in 2021 (Ministry for the Environment, 2023^[84]) all put a downward pressure on nutrient use, especially in the case of Nitrogen.

Figure 7. Trends in output, input and externalities (New Zealand - 1990-2021)

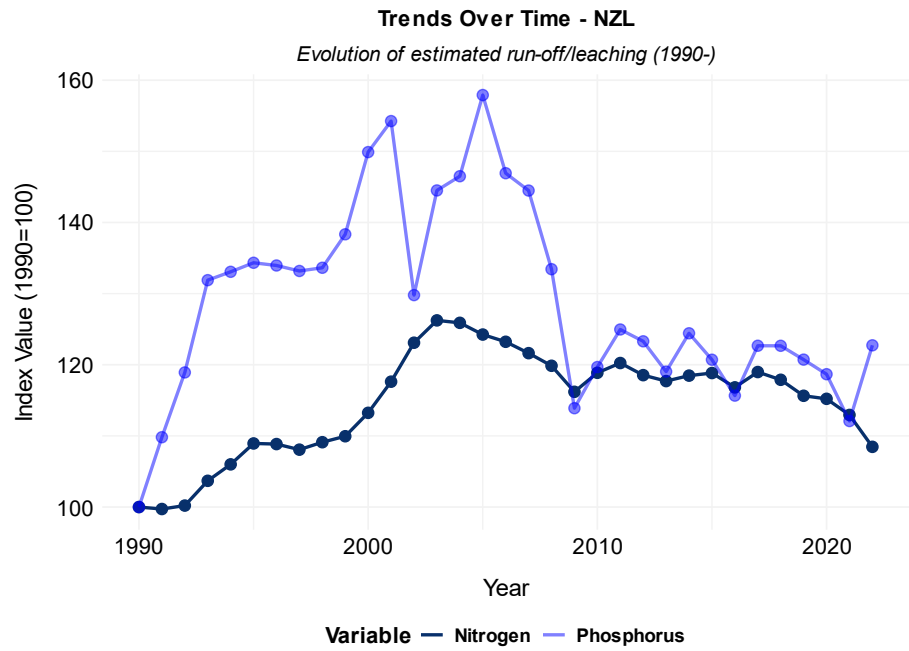
a. Output and inputs



b. GHG and ammonia



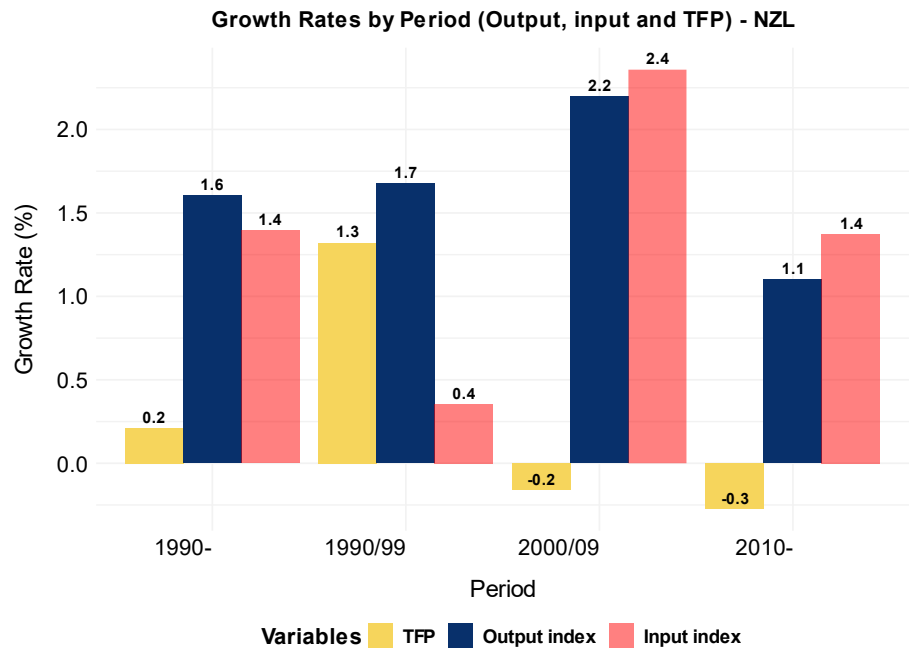
c. Estimated nutrient run-off and leaching



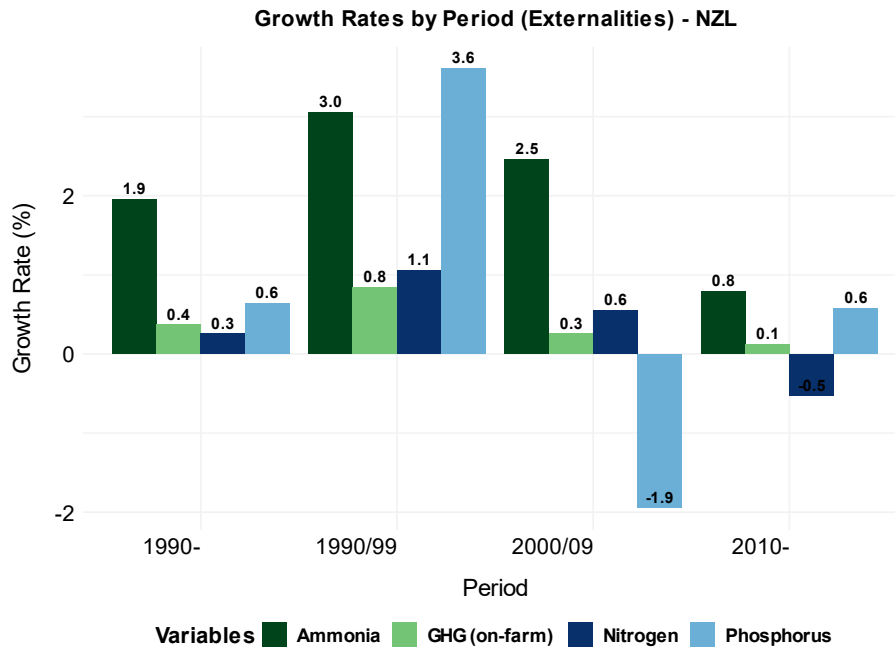
Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

Figure 8. Average growth rates by decade

a. Average TFP, output and input growth (by subperiod)



b. Average growth of externalities (by subperiod)



Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

SPG trends over time and decomposition

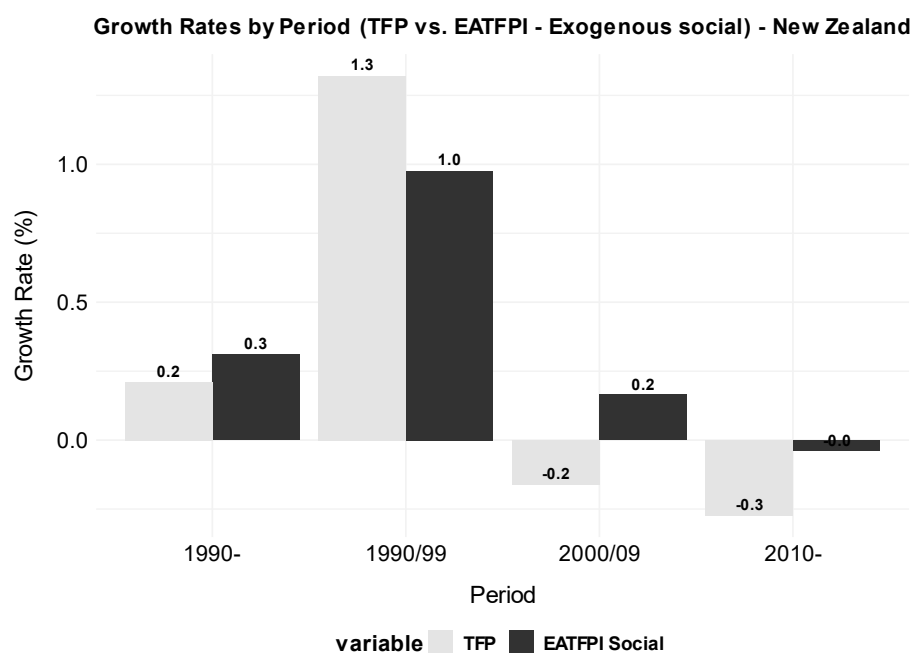
In the New Zealand case, adjusting TFP metrics to account for environmental externalities leads to a slight increase in productivity growth from 0.21% p.a. to 0.31% p.a. (Figure 9). The main reason why the two metrics (adjusted and unadjusted) have similar growth rates is that the economic value of the externalities increased at a similar pace as input use.

However, the small differences in the overall trend mask bigger differences across the different subperiods. During the 1990s, the large changes in composition of the livestock sector led to a situation where the growth rate of the environmental externalities outpaced that of input use, which led to a lower EATFPI value (1.0%) than traditional TFP (1.3%). This coincided with a period of sustained increase in the dairy herd sizes. However, for the next two decades, while traditional TFP metrics suggest a negative TFP growth rate, EAP metrics suggest a slightly positive one. This is attributable to the slowdown in the growth rate of the different externalities since 2000, as a result of efficiency gains, widespread adoption of management practices, as well as a stabilisation of herd sizes.

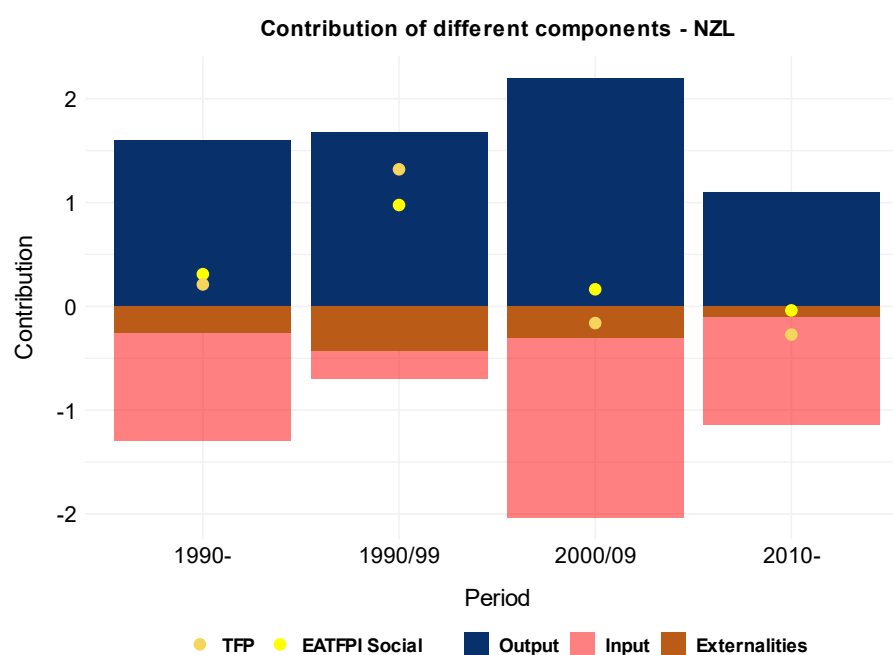
The contribution of externality growth was limited compared to that of input growth for all decades except the 1990s when the growth in externality values was higher than that of input growth (Figure 9, panel b). Within externalities, ammonia and GHG emissions had the highest contributions to the adjustment of the productivity metric (Figure 9, panel c). In the case of ammonia, it is an externality with a high social price that increased the most during the period.

Figure 9. TFP vs. EATFPI (Social) in New Zealand (by subperiod)

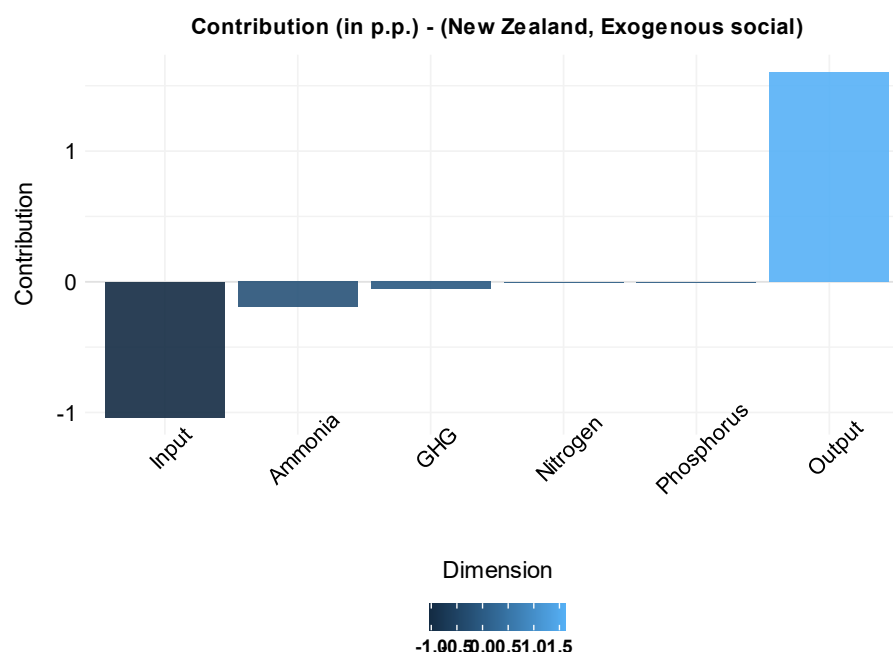
a. Average TFP and EATFPI growth rate (by subperiod)



b. Contribution to average EATFPI growth rate (by subperiod)



c. Contribution of individual externalities to average EATFPI (in p. p.)



Source: calculations based on USDA ERS (2024^[37]) and OECD (2025^[38]).

Limitations of the analysis

It is important to acknowledge limitations of the analysis, some of which are common to SPG metrics in general, and others specific to the country case study analysis.

One important aspect is that discussion in this paper focuses exclusively on growth rates, rather than levels. This is a limitation, as levels are highly relevant from a policy perspective. Cross-country comparison of TFP levels, even in the absence of environmental externalities, is also challenging. The database used in this paper (ERS IAP database) is non-axiomatic, which means that, while it allows for comparisons of growth rates, it does not allow for comparison of productivity levels. However, carrying out such analysis in contexts where relevant data allows for comparison of levels (e.g. analyses at sub-sectoral levels), while outside the scope of this paper, could be important from a policy perspective.

The focus on growth rates, rather than levels is a limitation that extends to environmentally adjusted productivity metrics, as these do not measure sustainability in an absolute sense. Growth rates of SPG metrics mostly measure the difference in growth rates between an environmentally adjusted output/input and a traditional input/output. This has four implications. First, estimated SPG does not necessarily imply that production is sustainable. Second, depending on the metric, it may not even imply that the absolute value of the externalities is decreasing. Third, even when metrics are improving, they may not be improving everywhere and there may be places where agriculture is becoming more unsustainable.⁴⁰ Lastly, because SPG is measured in growth terms, estimates can be sensitive to initial conditions. Traditional TFP growth tends to slow at higher output levels, while substantial relative reductions in environmental externalities are more likely to occur where initial levels are high. This highlights the need to interpret SPG estimates within their specific context in order to accurately capture these dynamics.

Therefore, while growth of environmentally SPG is a necessary condition for sustainability, it does not imply that production is sustainable. There are alternative frameworks that try to measure sustainability in a more absolute sense, such as the Sustainable Agriculture Matrix (SAM) approach (Zhang et al., 2021^[85]). The focus of these measures is not to measure integrated economic performance, but focus instead on

comparing raw values of economic, environmental and social variables against pre-defined sustainability thresholds, often based on expert opinion. The two approaches (SPG and focusing on absolute levels of externalities) are therefore complementary, but looking in-depth at sustainable production metrics is beyond the scope of this paper.

In this paper, the focus is on comparing how different methodological assumptions impact the estimated SPG for the case study countries. The aim is therefore to compare trends of SPG and how these compare to TFP growth rates within a country. This differs from what is generally done in many TFP analysis where the focus is cross-country comparisons (Salazar, Tadeo and Alvarez, 2024^[86]; Agnew and Nakelse, 2024^[87]; Fuglie, Morgan and Jelliffe, 2024^[2]). As a result, in order to reflect country-specificities the approach in this paper uses different prices to value environmental externalities across countries (to reflect different realities and abatement costs in different countries), which does not allow for cross-country comparisons of growth rates in this paper.⁴¹ Given the focus of the paper of temporal comparisons within a country, it is justified to use country-specific values. This also serves to highlight the differences in valuations of the different externalities obtained from the literature, as well as challenges to obtain comparable values across different countries (see Annex C, Table A C.3. for the assumptions used). However, importantly, having an indicator more fit-for-purpose for cross-country comparisons would require further harmonisation of methodologies and valuations of the environmental externalities.⁴²

Another aspect relates to important sources of uncertainty around valuation of environmental externalities. Econometric estimation of shadow prices can differ substantially depending on data, specification and method chosen (Luu and de Menna, 2025^[36]) and the same applies for external shadow prices, where typical ranges of environmental valuation are large. In the present analysis, given the number of specifications tested and the number of externalities included, values, rather than ranges were used. The main reason for this is that using ranges for each individual externality would create a very large number of potential combinations of values which could hamper interpretability of the results. The assumptions used in this paper are described in Annex C, but it is important to note the substantial uncertainty around shadow prices, especially when using valuations from a societal perspective. For GHG emissions, estimates of social cost of carbon have a large range, with estimates ranging from USD 40-525 per tonne of CO₂ equivalent (ToI, 2023^[88]) and some recent empirical analyses even use values above this range, with ranges from USD 252 to USD 928 per tonne of CO₂ equivalent used in a recent study (De Ridder and Lukasz, 2025^[41]). Similarly, in the EU context, for instance, estimates of damages related to ammonia emissions are accompanied by substantial uncertainty, with plausible ranges in the EUR 18 200 – EUR 39 500 per tonne range with similar uncertainty emerging for nitrogen and phosphorus (De Vries et al., 2024^[39]).

Another limitation relates to the choice of externalities included. In this paper, the approach was to include the same set of externalities across the studied countries and also to use, to the extent possible, context-relevant shadow prices for these externalities. However, it is important to highlight that not all externalities matter equally for all countries and sectors and having country- or sector-specific choices of externalities can make analysis more tailored to national contexts. In addition to this, the availability of regular and consistent data at national level constrains the choices of the dimensions used to adjust environmentally adjusted productivity measurements. In this analysis, for example, positive externalities are not included, nor are some important environmental dimensions (e.g. water in the case of Mexico). The reasons are twofold. First, for several externalities there are data gaps that preclude their inclusion for a yearly time series either because they may be calculated at irregular intervals (e.g. soil erosion) or because of the lack of datasets with comparable data sources across OECD countries (e.g. most positive externalities⁴³).

Finally, there are also uncertainties related to the estimate of run-off and leaching used. As explained in Annex C, the current approach relies on a constant fraction of leaching and run-off as a proportion of total inputs (often from National Inventory Reports in the case of Nitrogen and using a global average in the case of phosphorus). The fact that this fraction is assumed to be time-invariant means that it will not fully capture improvements in countries where this fraction has changed over time.⁴⁴

4 Conclusion

Several approaches exist to measure SPG. This analysis tested the importance of three methodological choices (weights, dimensions and indices) in the context of SPG metrics using three country case studies.

A key finding is that the weights assigned to the environment are central in explaining differences between TFP and SPG rates. When environmental weights are low (i.e. as a result of few dimensions or because unit values of externalities are lower), the SPG results closely track traditional TFP. This is intuitive as, in this case, growth is mostly driven by trends in outputs and inputs, rather than those of agri-environmental performance. Differences across indices become more pronounced as environmental weights increase, reflecting that some rely on partial decoupling between the externalities and output/input and others rely on absolute decoupling and require externalities to decrease at a certain pace.

Three case studies results illustrate this pattern. Across all countries, differences between SPG and TFP were generally small when valuing single externalities at producer prices. However, they became larger as weights increased. For all countries, SPG rates tended to exceed TFP growth rates when all externalities were accounted using indices that rely on partial decoupling. However, in the cases of Mexico and New Zealand, this was not the case when using indices that rely of absolute decoupling. This should not be interpreted as poor performance, as achieving absolute decoupling alongside cumulative output expansions exceeding 50% over three decades would be extremely challenging.

While the methodology of the paper does not allow for a formal causal attribution of SPG rates to specific policies and technologies, the decomposition of SPG also showed that potential drivers of SPG are country-specific. In the cases of Mexico and New Zealand, trade policies, high SPG rates coincide with periods of profound reforms of agricultural support systems, as well as the adoption of technologies and practices. In the case of the Netherlands, high SPG rates coincide with regulations (especially ammonia, manure and fertiliser) which were identified as potential key precursors of large reductions in several of the externalities.

Finally, the paper serves to illustrate that there are different ways to compute SPG metrics that can help interpret SPG performance at country level. Under certain conditions, the choice of index will have a limited practical impact on the estimated growth rates. The way in which externalities are valued and the dimensions included, on the other hand, have a consistently high impact on the SPG rates. It is therefore important to carry out sensitivity checks of these methodological choices. The most appropriate methodology is thus likely to depend on the intended use of the SPG measure, its intended interpretation, and the desired perspective of the valuation and dimensions to be included.

References

- Acosta, A., C. Barrantes and R. Ihle (2020), “Animal disease outbreaks and food market price dynamics: Evidence from regime-dependent modelling and connected scatterplots”, *Australian Journal of Agricultural and Resource Economics*, Vol. 64/3, pp. 960-976, <https://doi.org/10.1111/1467-8489.12387>. [53]
- Anderson, K. and A. Valdés (2007), “Distortions to Agricultural Incentives in Latin America”. [44]
- Ang, F., H. Dakpo and S. Pieralli (2022), “Insights into the Measurement of Agricultural Total Factor Productivity and the Environment”, in *Insights into the Measurement of Agricultural Total Factor Productivity and the Environment*, <https://www.oecd.org/agriculture/topics/network-agricultural-productivity-and-environment/>. [14]
- Backus, G. (2017), *An Overview and Assessment of Policy Instruments in the Netherlands*, <http://www.worldbank.org>. [61]
- Baldoni, E. et al. (2025), *Environmentally Adjusted TFP in EU Agriculture - Insights from a methodological proposal*, [https://www.oecd.org/content/dam/oecd/en/networks/network-on-agricultural-total-factor-productivity-and-the-environment/2025-tenth-meeting-network-agricultural-tfp-and-environment.zip/ jcr_content/renditions/original.media_file.download_attachment.file/2025-tenth-meeting-network-agricultural-tfp-and-environment.zip](https://www.oecd.org/content/dam/oecd/en/networks/network-on-agricultural-total-factor-productivity-and-the-environment/2025-tenth-meeting-network-agricultural-tfp-and-environment.zip/jcr_content/renditions/original.media_file.download_attachment.file/2025-tenth-meeting-network-agricultural-tfp-and-environment.zip) (accessed on 13 April 2026). [13]
- Ball, E. et al. (2001), “Productivity of the US Agricultural Sector: The Case of Undesirable Outputs”, in Hulten, C., E. Dean and M. Harper (eds.), *New developments in productivity analysis*, University of Chicago Press. [33]
- Berkhout, P., H. Meulen and P. Ramaekers (2023), *Staat van Landbouw, Natuur en Voedsel : editie 2023*, Wageningen Economic Research, <https://doi.org/10.18174/641747>. [67]
- Bouwman, A., A. Beusen and G. Billen (2009), “Human alteration of the global nitrogen and phosphorus soil balances for the period 1970–2050”, *Global Biogeochemical Cycles*, Vol. 23/4, <https://doi.org/10.1029/2009GB003576>. [92]
- Bureau, J. and J. Antón (2022), “Agricultural Total Factor Productivity and the environment: A guide to emerging best practices in measurement”, *OECD Food, Agriculture and Fisheries Papers*, No. 177, OECD Publishing, Paris, <https://doi.org/10.1787/6fe2f9e0-en>. [12]
- Cárdenas Rodríguez, M., I. Haščíč and M. Souchier (2018), “Environmentally Adjusted Multifactor Productivity: Methodology and Empirical Results for OECD and G20 Countries”, *Ecological Economics*, Vol. 153, pp. 147-160, <https://doi.org/10.1016/j.ecolecon.2018.06.015>. [10]

- Cárdenas Rodríguez, M. et al. (2023), “Environmentally adjusted multifactor productivity: Accounting for renewable natural resources and ecosystem services”, *OECD Green Growth Papers*, No. 2023/01, OECD Publishing, Paris, <https://doi.org/10.1787/9096211d-en>. [9]
- Chancellor, W. et al. (2021), “Controlling for the effects of climate on total factor productivity: A case study of Australian farms”, *Food Policy*, Vol. 102, <https://doi.org/10.1016/j.foodpol.2021.102091>. [16]
- Chapman, D. et al. (2025), *Pasture productivity in New Zealand 1990–2020: trends, expectations, and key factors*, Taylor and Francis Ltd., <https://doi.org/10.1080/00288233.2024.2425071>. [72]
- Cobourn, K. et al. (2024), “An Index Theory Based Approach to Measuring the Environmentally Sustainable Productivity Performance of Agriculture”, *OECD Food, Agriculture and Fisheries Papers*, No. 213, OECD Publishing, Paris, <https://doi.org/10.1787/bf68fb78-en>. [25]
- De Ridder, M. and R. Lukasz (2025), *Adjusting productivity for carbon emissions: A new perspective on the growth slowdown*. [41]
- De Vries, J. et al. (2024), *Environmental Prices Handbook 2024: EU27 version Methodical justification of key indicators used for the valuation of emissions and the environmental impact*, <http://www.ce.nl>. [39]
- FAO (2025), *FAOSTAT Statistical Database: Livestock Patterns*. [49]
- FAO (2025), *World Census of Agriculture. FAOSTAT Database. Retrieved from* <https://www.fao.org/faostat>. [59]
- FAO (2024), *Greenhouse gas emissions from agrifood systems: Global, regional and country trends, 2000–2022*. [6]
- Färe, R. et al. (1993), “Derivation of Shadow Prices for Undesirable Outputs: A Distance Function Approach”, *The review of economics and statistics*, Vol. 75/2, pp. 374-380, <https://doi.org/10.2307/2109448>. [19]
- Flett, R. et al. (2004), “The technology acceptance model and use of technology in New Zealand dairy farming”, *Agricultural Systems*, Vol. 80/2, pp. 199-211, <https://doi.org/10.1016/j.agsy.2003.08.002>. [79]
- Fraters, B. et al. (2004), “Agricultural practice and water quality in the Netherlands in the 1992-2002 period. Background information for the third EU Nitrates Directive Member States report”, No. Report no. 500003002/2004. [62]
- Fuglie, K. (2015), “Accounting for growth in global agriculture”, *Bio-based and Applied Economics*, Vol. 4/3, pp. 201-234, <https://doi.org/10.13128/BAE-17151>. [89]
- Fuglie, K. et al. (2020), *Water, Policy, and Agriculture Productivity in Egypt*. [18]
- Fuglie, K. et al. (2020), “Water, Policy, and Productivity in Egyptian Agriculture”, *American Journal of Agricultural Economics*, Vol. 103/4, pp. 1378-1397, <https://doi.org/10.1111/ajae.12148>. [31]

- Fuglie, K., S. Morgan and J. Jelliffe (2024), *World Agricultural Production, Resource Use, and Productivity, 1961-2020*, <http://www.ers.usda.gov>. [2]
- Ge, H. (2025), "An Overview of New Zealand's Economic and Environmental Sustainability in Agriculture: 40 Years Without Subsidies", *Journal of Sustainability*, Vol. 1/1. [69]
- Gobierno de México (2025), *Mexico: Informe del Inventario Nacional de Emisiones de gases y compuestos de efecto Invernadero, 1990-2022*. [50]
- Gruère, G., M. Shigemitsu and S. Crawford (2020), "Agriculture and water policy changes: Stocktaking and alignment with OECD and G20 recommendations", *OECD Food, Agriculture and Fisheries Papers*, No. 144, OECD Publishing, Paris, <https://doi.org/10.1787/f35e64af-en>. [3]
- Gu, W., J. Hussain and M. Willox (2019), "Environmentally Adjusted Multifactor Productivity Growth for the Canadian Manufacturing Sector", *Analytical Studies Branch Research Paper Series*, <http://www.statcan.gc.ca>. [15]
- Harris, B. (2005), "Breeding dairy cows for the future in New Zealand", *New Zealand Veterinary Journal*, Vol. 53/6, pp. 384-389, <https://doi.org/10.1080/00480169.2005.36582>. [73]
- Henderson, B. and J. Lankoski (2023), "Integrated approaches for agricultural sustainability and productivity assessments", *OECD Food, Agriculture and Fisheries Papers*, No. 204, OECD Publishing, Paris, <https://doi.org/10.1787/60cfa586-en>. [26]
- Henningesen, A. (2025), "2.4. Proposal for a Sustainable Total Factor Productivity metric using societal shadow values", in OECD (ed.), *Sustainable Agricultural Productivity to Address Food Systems Challenges: Measurement, Data, Drivers and Policies, Proceedings from the OECD Conference, 28 October 2024*. [20]
- Hergoualch, K. et al. (2019), *Chapter 11: N2O Emissions from Managed Soils, and CO2 Emissions from Lime and Urea Application N2O Emissions from Managed Soils, and CO2 Emissions from Lime and Urea Application*. [94]
- Hugonnet, M. (2025), "Liberalisation of agricultural policies: the case of New Zealand", *Ministry of Agriculture and Food Sovereignty - Analysis note*, No. 210, <https://doi.org/10.1023/B:GEJO.0000019969.38496.82>. [68]
- IFDC (2008), "IFDC report". [81]
- Jongeneel, R., N. Polman and G. Cornelis Van Kooten (2016), *How Important are Agricultural Externalities? A Framework for Analysis and Application to Dutch Agriculture*, <http://web.uvic.ca/~repa/>. [28]
- Kempenaar, C. (2020), *Precision farming on Dutch arable farms: status, challenges and outlook*. [65]
- Lord, S. (2023), *Data for Background Report 2023 - FAO State of Food and Agriculture*. [40]
- Luu, N. and B. de Menna (2025), *Mind the gap: Accounting for Emissions in Productivity Measurement*. [36]

- Macintosh, K., R. McDowell and C. Thiange (2025), “A 10-year evaluation of management practices and nutrient losses from dairy farms in New Zealand – Trends and drivers”, *Agriculture, Ecosystems and Environment*, Vol. 377, <https://doi.org/10.1016/j.agee.2024.109261>. [83]
- Mckenzie World Bank, D. (2006), “The Consumer Response to the Mexican Peso Crisis”, *Economic Development and Cultural Change*, Vol. 55/1, <https://doi.org/10.1086/505721>. [46]
- Ministry for the Environment (2025), *New Zealand’s Greenhouse Gas Inventory 1990–2023*, Ministry for the Environment, Wellington. [70]
- Ministry for the Environment (2023), *Synthetic nitrogen fertiliser cap: implementation guidance on Essential Freshwater policies and regulations*. [84]
- Ministry for the Environment and Ministry for Primary Industries (2021), *The government response to the findings of the Overseer peer review report*, Ministry for the Environment and Ministry for Primary Industries, Wellington. [78]
- Murray-Tortarolo, G. and V. Jaramillo (2019), “The impact of extreme weather events on livestock populations: the case of the 2011 drought in Mexico”, *Climatic Change*, Vol. 153/1-2, pp. 79-89, <https://doi.org/10.1007/s10584-019-02373-1>. [52]
- Murty, S., R. Robert Russell and S. Levkoff (2012), “On modeling pollution-generating technologies”, *Journal of Environmental Economics and Management*, Vol. 64/1, pp. 117-135, <https://doi.org/10.1016/j.jeem.2012.02.005>. [21]
- Nanere, M. et al. (2007), “Environmentally adjusted productivity measurement: An Australian case study”, *Journal of Environmental Management*, Vol. 85/2, pp. 350-362, <https://doi.org/10.1016/j.jenvman.2006.10.004>. [32]
- Nanere, M. et al. (2007), “Environmentally adjusted productivity measurement: An Australian case study”, *Journal of Environmental Management*, Vol. 85/2, pp. 350-362, <https://doi.org/10.1016/j.jenvman.2006.10.004>. [17]
- Nin-Pratt, A. (2025), “Reassessing agricultural growth: Introducing a sustainable productivity index for Latin America and the Caribbean”, in Salazar, L. et al. (eds.), *Agricultural Productivity in Latin America and the Caribbean: What we know and where are we heading*, IDB, Washington, DC. [35]
- Njuki, E. and C. O'Donnell (2025), “Sustainable Productivity Change in the U.S. Dairy Sector”, *CEPA Working Papers Series WP012025*, <https://ideas.repec.org/p/qld/uqcepa/195.html>. [24]
- O'Donnell, C. (2022), “How to Build Sustainable Productivity Indexes”, *CEPA Working Papers Series WP102022*, <https://ideas.repec.org/p/qld/uqcepa/182.html>. [23]
- O'Donnell, C. (2018), *Productivity and Efficiency Analysis. An Economic Approach to Measuring and Explaining Managerial Performance*, <https://doi.org/10.1007/978-981-13-2984-5>. [22]
- OECD (2025), *Agri-environmental Indicators Database*. [38]
- OECD (2025), *Environmental Performance of Agriculture in OECD Countries: Key Trends and Insights*, OECD Publishing, Paris, <https://doi.org/10.1787/2679ba38-en>. [7]

- OECD (2024), *Agricultural Policy Monitoring and Evaluation 2024: Innovation for Sustainable Productivity Growth*, OECD Publishing, Paris, <https://doi.org/10.1787/74da57ed-en>. [43]
- OECD (2024), *The challenge of measuring and achieving sustainable agricultural productivity growth*, https://www.oecd.org/en/publications/the-challenge-of-measuring-and-achieving-sustainable-agricultural-productivity-growth_1c17e895-en.html. [8]
- OECD (2023), *Policies for the Future of Farming and Food in the Netherlands*, OECD Agriculture and Food Policy Reviews, OECD Publishing, Paris, <https://doi.org/10.1787/bb16dea4-en>. [56]
- OECD (2022), *Insights into the Measurement of Agricultural Total Factor Productivity and the Environment*, <https://www.oecd.org/agriculture/topics/network-agricultural-productivity-and-environment/>. [11]
- OECD (2021), *Making Better Policies for Food Systems*, OECD Publishing, Paris, <https://doi.org/10.1787/ddfba4de-en>. [1]
- OECD (2019), *Agriculture and the environment*, <https://www.oecd.org/en/topics/agriculture-and-environment.html>. [4]
- OECD (2015), *Innovation, Agricultural Productivity and Sustainability in the Netherlands*, OECD Food and Agricultural Reviews, OECD Publishing, Paris, <https://doi.org/10.1787/9789264238473-en>. [57]
- Ortega Diaz, A. and A. Guerrero (2025), “Mexico”, in Salazar, L. et al. (eds.), *Agricultural Productivity in Latin America and the Caribbean: What we know and where are we heading*, IDB, Washington, DC. [42]
- Ortega-Gaucin, D., J. Bartolón and H. Bahena (2018), “Drought vulnerability indices in Mexico”, *Water (Switzerland)*, Vol. 10/11, <https://doi.org/10.3390/w10111671>. [51]
- Ortiz, R. et al. (2008), *Wheat genetic resources enhancement by the International Maize and Wheat Improvement Center (CIMMYT)*, <https://doi.org/10.1007/s10722-008-9372-4>. [48]
- Pretty, J. et al. (2000), “An assessment of the total external costs of UK agriculture”, *Agricultural systems* 65, pp. 113-136, [https://doi.org/10.1016/S0308-521X\(00\)00031-7](https://doi.org/10.1016/S0308-521X(00)00031-7). [27]
- Prins, H. et al. (2018), *Damage to Dutch agricultural and horticultural crops as a result of the drought in 2018*. [66]
- Puyana, A. (2012), “Give to AgEcon Search Mexican Agriculture and NAFTA: A 20-Year Balance Sheet”, *Review of agrarian studies*, Vol. 2/1, <http://ageconsearch.umn.edu>. [47]
- Quinn, J. et al. (2009), “Grassland farming and water quality in New Zealand”, *Tearmann: Irish journal of agri-environmental research*, Vol. 7, pp. 69-88, <https://www.researchgate.net/publication/261028008>. [80]
- Rijk, B., M. van Ittersum and J. Withagen (2013), “Genetic progress in Dutch crop yields”, *Field Crops Research*, Vol. 149, pp. 262-268, <https://doi.org/10.1016/j.fcr.2013.05.008>. [58]
- RIVM (2025), “Methodology for the calculation of emissions from agriculture”, No. RIVM-2025-0003. [91]
- RIVM (2024), “Agricultural practices and water quality in the Netherlands: status (2020-2023) and trends (1992-2023)”, *RIVM report*, No. 0209. [64]

- RIVM (2016), *ReCiPe 2016 A harmonized life cycle impact assessment method at midpoint and endpoint level Report I: Characterization*. [93]
- Ros, J. (2001), "From the Capital Surge to the Financial Crisis and Beyond: The Mexican Economy in the 1990s", in Ffrench-Davis, R. (ed.), *Financial crises in "successful" emerging economies*. [45]
- Salazar, L., D. Tadeo and L. Alvarez (2024), *Agricultural Productivity in the Latin America and Caribbean Region (1961-2021)*, <http://www.iadb.org>. [86]
- Salinger, M. and A. Porteous (2014), "New Zealand climate: patterns of drought 1941/42-2012/13", *Weather and Climate*, Vol. 34, pp. 2-19, <https://www.researchgate.net/publication/267570085>. [74]
- Selbie, D. et al. (2013), *Understanding the distribution and fate of nitrogen and phosphorus in OVERSEER®*, <http://www.overseer.org.nz>. [90]
- Skolrud, T. et al. (2020), *Measuring Externalities in Canadian Agriculture: Understanding the Impact of Agricultural Production on the Environment*, <http://www.capi-icpa.ca>. [29]
- Sparling, G. and L. Schipper (2004), "Soil quality monitoring in New Zealand: Trends and issues arising from a broad-scale survey", *Agriculture, Ecosystems and Environment*, Vol. 104/3, pp. 545-552, <https://doi.org/10.1016/j.agee.2003.11.014>. [76]
- Sparling, G. et al. (2004), "Soil quality monitoring in New Zealand: Practical lessons from a 6-year trial", *Agriculture, Ecosystems and Environment*, Vol. 104/3, pp. 523-534, <https://doi.org/10.1016/j.agee.2004.01.021>. [75]
- StatsNZ (2024), *Livestock numbers: Data to 2023*. [71]
- Stegeman, A. et al. (2004), "Avian Influenza A Virus (H7N7) Epidemic in the Netherlands in 2003: Course of the Epidemic and Effectiveness of Control Measures", *The Journal of Infectious Diseases*, Vol. 190/12, pp. 2088-2095, <https://academic.oup.com/jid/article/190/12/2088/862959>. [63]
- Thompson, T. (ed.) (2024), *Global agricultural productivity report - Scaling high impact bundles of proven and emerging tools*, <http://www.globalagriculturalproductivity.org>. [87]
- Tol, R. (2023), "Social cost of carbon estimates have increased over time", *Nature Climate Change*, Vol. 13/6, pp. 532-536, <https://doi.org/10.1038/s41558-023-01680-x>. [88]
- Torres Castillo, N. et al. (2020), *Impact of climate change and early development of coffee rust – An overview of control strategies to preserve organic cultivars in Mexico*, Elsevier B.V., <https://doi.org/10.1016/j.scitotenv.2020.140225>. [54]
- Tubiello, F. et al. (2021), "Greenhouse gas emissions from food systems: Building the evidence base", *Environmental Research Letters*, Vol. 16/6, <https://doi.org/10.1088/1748-9326/ac018e>. [5]
- U.S. Department of Agriculture, E. (2024), *International Agricultural Productivity data product*. [37]
- van der Net, L. et al. (2024), "Greenhouse gas emissions in the Netherlands 1990–2022. National Inventory Report 2024", *RIVM report*, No. 2024-0017, <https://doi.org/10.21945/RIVM-2024-0017>. [60]

- Verevis, S. and M. Üngör (2021), “What has New Zealand gained from The FTA with China?: Two counterfactual analyses†”, *Scottish Journal of Political Economy*, Vol. 68/1, pp. 20-50, <https://doi.org/10.1111/sjpe.12260>. [77]
- Vrachioli, M. and V. Tzouvelekas (2022), “Non-commodity (bad) outputs: Water quality”, in Bureau, J. (ed.), *Insights into the Measurement of Agricultural Total Factor Productivity and the Environment*. [30]
- Wang, A., S. Hussain and J. Yan (2024), “Evaluating the interlinkage between pesticide residue regulation and agricultural green total factor productivity: empirical insights derived from the threshold effect model”, *Environment, Development and Sustainability*, <https://doi.org/10.1007/s10668-024-04744-w>. [34]
- World Bank (2025), *World Bank Commodities Price data (The Pink Sheet)*. [55]
- Yang, W. et al. (2021), “Analysis of adoption trends of in-parlor technologies over a 10-year period for labor saving and data capture on pasture-based dairy farms”, *Journal of Dairy Science*, Vol. 104/1, pp. 431-442, <https://doi.org/10.3168/jds.2020-18726>. [82]
- Zhang, X. et al. (2021), “Quantitative assessment of agricultural sustainability reveals divergent priorities among nations”, *One Earth*, Vol. 4/9, pp. 1262-1277, <https://doi.org/10.1016/j.oneear.2021.08.015>. [85]

Notes

¹ While there are empirical differences between productivity indicators and indices (e.g. indices are typically ratio measures whereas indicators are measures of net output), the term index will be used throughout the paper to describe the different ways of measuring SPG. The six indices considered are: environmentally adjusted TFP indicators with input (EATFP-I) and output (EATFP-O) adjustments, the environmentally sustainable productivity index with input (ESPII) and output (ESPIO) adjustments, and the weak environment-productivity growth indicator where the weights of the externality are given by their importance relative to the value of inputs (WEPI) and outputs (WEPO).

² The term Sustainable Productivity Growth, in the context of this paper, refers to the growth rate of indices that measure integrated agri-environmental and productivity performance by combining traditional Total Factor Productivity (TFP) with environmental variables or externalities. Importantly, the metrics discussed in this paper do not measure sustainability in an absolute sense (i.e. they do not measure environmental variables against a given threshold that below which performance is considered to be environmentally sustainable).

³ From here onwards the acronym GHG emissions will refer to direct agricultural emissions of carbon dioxide (CO₂), nitrous oxide (N₂O) and methane (CH₄), consistent with the [OECD Agri-Environmental Indicators](#).

⁴ Case study countries were selected to include countries in different continents, with different production systems, policies, and where trends in traditional TFP growth have differed, with the aim of increasing the likelihood that features of specific indicators and assumptions that apply to countries with specific TFP growth are captured.

⁵ There are additional methodological choices involved when developing an environmentally adjusted productivity metrics, such as the use of superlative vs. non-superlative indices.

⁶ The indicators highlighted above were selected as they reflect existing work on the topic at the OECD and in its networks when estimating sustainable productivity indicators at the macro-level.

⁷ The main reason why producer-centric valuations of externalities are more methodologically consistent is that both traditional outputs and inputs are valued at prices faced by the producers. Therefore, valuing externalities at prices/costs faced by producers is more methodologically consistent as all inputs, outputs and externalities are being valued using the same perspective. If an adjustment was made based on societal prices, this would require adjusting all inputs and outputs. The counterargument often used for society-centric valuations is that producer-centric valuations are typically too low (as they do not reflect the

cost to society) and that market prices of traditional outputs and inputs are often a good approximation of their societal prices.

⁸ The executive summary and key messages refer to six indicators because two versions of the Weak Environment Productivity growth indicator are calculated. One where weights are based on the value of the externalities relative to output and another where value of externalities is calculated relative to inputs.

⁹ In the context of the International Agricultural Productivity database (U.S. Department of Agriculture, 2024^[37]), while time-invariant prices are assumed in the numerator in the components that rely on FAO's output index, in the case of inputs, the database relies on time-varying cost-shares based on empirical estimates for several countries, which capture some of the effect of relative prices.

¹⁰ Given that externalities do not typically have market prices, the price of externalities is commonly referred to as a shadow price. In the context of joint production of good and bad outputs, it can be thought of as the opportunity cost of reducing a negative externality by one unit (e.g. one example would be the foregone production/profit necessary to reduce a given negative externality). In practice, there are different ways of obtaining and estimating shadow prices.

¹¹ One problematic property of such an indicator is that it is possible to have negative environmentally adjusted output values when the monetary valuations of the externalities exceed the value of production.

¹² A shadow price essentially measures the marginal change in value that stems from changing a constraint. In the case of SPG metrics, it typically refers to the opportunity cost—generally measured in the form of foregone output, revenue or profit—of reducing a given externality.

¹³ In settings where countries are being compared, this limit is typically set by the country with the highest economic value of externalities relative to output.

¹⁴ In the original paper, the common base value used is the OECD average for 1990 (Cobourn et al., 2024^[25]).

¹⁵ A value of 1 implies that policymakers only care about the environmental externality. A value of 0 implies that policymakers only give importance to traditional TFP. Increasing values of the parameter indicate an increasing importance of policymakers given to the externality.

¹⁶ In the original paper, the authors use TFP growth rate and the growth rate in emission intensity from agriculture using equal weights (Henderson and Lankoski, 2023^[26]). The authors also propose a second metrics, the Strong Environment-Productivity growth indicator (SEP), that is based on a strong environmental sustainability rule which takes the growth rate of the worst performing category. This measure was not kept as indicator to compare against because the weighing scheme is not comparable with that of other SPG metrics. Moreover, the assumption in the SEP of using the lowest growth rate across all the measures also implies that this metric is sensitive to the inclusion of many dimensions.

¹⁷ The letter gamma is used instead of alpha to avoid implying that the weight attributed to the environmental externalities is set in a similar way to that of ESPI. Gamma is thus used to denote environmental weights in a more general way. It will be equal to alpha in indicators where an exogenous parameter is used (e.g. ESPI), but not necessarily in other cases.

¹⁸ Importantly, while the conceptual equivalence of gamma means that gamma has the same conceptual interpretation, it does not necessarily mean that the relative importance of the environmental externalities

(i.e. the ratio of the marginal effects) is equivalent in all cases. For example, this ratio is likely to differ across metrics that adjust inputs vs. metrics that adjust outputs. There is therefore a higher degree of comparability within metrics that adjust inputs or outputs rather than between metrics that adjust inputs and outputs, partially because conceptually *gamma* has a different meaning depending on whether inputs or outputs are being adjusted. In this paper, different *gammas* are used for the EATFPI and EATFPO as a starting point to calibrate the *gammas* of ESPII and ESPIO, respectively. Having a unique value of *gamma* across indicators (e.g. by assuming a unique weight, such as $\frac{\frac{\partial I}{\partial EI}}{\frac{\partial I}{\partial EI} + \frac{\partial I}{\partial TFP}}$ in such a way that the relative importance of the environment vis-à-vis TFP is constant across all indicators) would ensure a more exact numerical comparability of all indicators across time. However, achieving this between EATFPI and EATFPO would require time-varying unit prices of externalities (when other unit prices are constant). In terms of the comparability of the weights between the EATFPI and ESPII and the EATFPO and ESPIO pairs, setting the weight of *alpha* equal to *gamma* already ensures that the condition $\frac{\frac{\partial I}{\partial EI}}{\frac{\partial I}{\partial EI} + \frac{\partial I}{\partial TFP}}$ is met at the base period and weights will be similar for small growth rates of YI, EI and XI. However, ensuring exact numerical equivalence of the weights for all periods would imply setting a time-varying value of *alpha* which would contradict the original formulation of the indicators which relies on a fixed value of *alpha*.

¹⁹ With a weight of 0.5 and assuming an increase in the level of the externality, this essentially means that the growth rate of TFP needs to be at least as large as the growth rate of the externality.

²⁰ In Pretty et al. (2000^[27]) the results also highlight the importance of damages to human health as a result of animal health outbreaks and pesticides in the water. In Canada, Skolrud et al. (2020^[29]) highlight soil erosion as an externality with a high environmental cost.

²¹ The term external prices in this context refers to the fact that these costs come from the literature not that abatement costs themselves are exogenous or external.

²² While endogenous prices are producer-centric, external prices can either be producer-centric (e.g. producer abatement costs) or be based on a societal perspective (e.g. estimates based on damage functions).

²³ On-farm emissions consist of carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O).

²⁴ It is possible for shadow prices to not be statistically different from 0, in which case the externality has a weight of 0 in the environmentally adjusted metric. In cases where the incentive structure that farmers implies that the costs of the externalities are not internalised, this may result in very low shadow prices, even when the externality is known to matter from a policy perspective.

²⁵ For multiple externalities, prices are likely to be sourced from different sources using different methodologies.

²⁶ It could be argued that it is the only adjustment that can be done in a methodologically consistent way as the societal prices of other inputs/outputs are not observed.

²⁷ A second argument is that for traditional outputs and inputs, the social prices should be approximately equal to their market prices.

²⁸ The difference in the WEP indicators (WEPI and WEPO) is that in WEPI relates to the different choice of γ . WEPI uses the same value of gamma as in ESPII whereas WEPO uses the same value of γ as in ESPIO.

²⁹ For the input adjustment, since the starting point is the USDA input index, adjusting the input value requires an assumption on the value of inputs. In this paper, the value of inputs is assumed to be equal to the value of outputs 2015, which allows us to adjust the denominator using the value of externalities. No such adjustment is required for the output adjustment.

³⁰ Given structural differences in the agricultural sector of the case study countries, the composition of externalities will differ. As a result, within-country comparisons of the traditional TFP with metrics adjusted for all externalities at the same time are most meaningful. This being said, adjustments for individual externalities are kept in the results tables to show the effect of individual externalities on the adjusted growth rates.

³¹ For the Netherlands and New Zealand, the ESPII and ESPIO metrics need to be compared to a slightly different growth rate of traditional TFP. Specifically, they need to be compared to 0.36% (Netherlands) and 0.19% (New Zealand). The reason is that ESPI metrics have a constant alpha whereas the USDA cost-shares change over time for those two countries. This means that the exact same growth rates are obtained for every year, but there is a difference in the year (2001) where cost shares change.

³² According to FAOSTAT, production of vegetables increased by 53% between 1990-2000, primary fruits by approximately 45%, milk by 51% and poultry meat by 133%. Over the same time, cereal production increased by 5%.

³³ FAOSTAT data indicates a reduction 32% and 64% decrease in nutrient use between 1995 and 1985. Intensity of use of fertilisers started increasing again from 1995 onwards.

³⁴ Analysis stops in 2019 as it is the last year for which data on all four externalities are available.

³⁵ Exports of manure decrease the pressure on domestic nutrient balance but increase the pressures on importing countries.

³⁶ Values between 2019 and 2022 are interpolated as they were not available from Eurostat for the most recent version of the AEI dataset.

³⁷ According to FAOSTAT, these two groups of products account for approximately 90% of the increases in the value of production observed between 1990-2021.

³⁸ Comparing FAOSTAT data on dairy exports expressed in milk equivalent suggests that the entirety of additional milk production since 1990 was exported and suggests that 85% of milk production was exported in 2021.

³⁹ The NPS-FM was since amended in 2014, 2017 and 2020.

⁴⁰ This is a limitation in cases, such as this analysis, where the analysis is done at the national level where there may be substantial heterogeneity within a country. For example, in the case of Mexico, the sector is characterized by a combination of a large number of small-scale, often subsistence, farmers and a smaller

group of large-scale farms operating with cutting-edge technology, with very different levels of productivity (Ortega Diaz and Guerrero, 2025^[42]).

⁴¹ Note, however, that this aspect can be addressed, for instance, by ensuring constant relative prices that are common to all countries being analysed. Given the focus on temporal comparisons within a country, however, this was not pursued in this paper.

⁴² Ideally, to ensure full consistency with how the output index is constructed a unique price per externality and for individual inputs would facilitate cross-country comparisons. Importantly, however, while this is not done in this report, a common price for the environmental externalities could be computed using an aggregation of individual country-specific valuations.

⁴³ In a case study for the Netherlands (Jongeneel, Polman and Cornelis Van Kooten, 2016^[28]), the authors value positive externalities (care farming, agro-tourism, values related to biodiversity and landscape). The authors estimate that in value added terms, these account for approximate 2.5% of total benefits. However, databases containing similar data for many countries are not available.

⁴⁴ For the countries in the case study, however, while they have changed for the Netherlands, the changes have typically been relatively small (e.g. ranges during the period include 13-15%). In cases such as New Zealand, however, this approach is likely not to capture investments at farm-level in techniques such as dairy effluent management systems unless these have been explicitly captured in the withdrawals of nutrient balances.

Annex A. Derivations to express all indicators as a function of an environmental weight (γ)

An important aspect to bear in mind is that, while the valuation of the environment does not have an explicit parameter across all different metrics used to measure SPG, all metrics assign a weight to the environment. In a sense, especially in the case of constant prices all metrics can be expressed as a function of such a parameter, which will be called *gamma* in this paper.

Starting with traditional EATFP metrics, these metrics rely on the concepts of environmentally adjusted output or inputs to adjust traditional TFP metrics. Typically, they rely on augmenting the inputs (in the case of an input adjustment) or deducting the externalities from the value of output (in the case of an output adjustment).

EATFP – Output adjustment

EATFP metrics adjusted via the output, SPG (or EATFP) is given by the difference in the growth rates of environmentally adjusted output and traditional inputs. Importantly, while there is typically no explicit value for γ in the numerator, in such frameworks the relative importance of the environmental dimensions is determined by the relative magnitudes of externalities and traditional outputs.

For the discussion that follows, one output, one externality and one input are assumed, and prices are kept constant over time for simplicity. Defining:

Y_t - Quantity of agricultural output (excluding aquaculture) at time t

E_t - Quantity of a given environmental externality at time t

YI_t - Index value of agricultural output (excluding aquaculture) at time t

EI_t - Index value of environmental externality at time t

p_e - Time-invariant unit price of the environmental externality E

p_y - Time-invariant unit price of output Y

Environmentally adjusted output can be defined as:

$$EAO_t = p_y Y_t - p_e E_t \quad (1)$$

A normalised version of Equation (1) can be re-written as:

$$EAO_t = (p_y + p_e) * \left(\left(\frac{p_y}{p_y + p_e} \right) * Y_t - \left(\frac{p_e}{p_y + p_e} \right) * E_t \right) \quad (2)$$

Where the relative importance of the traditional output and the environmental externality are given by the terms $\left(\frac{p_y}{p_y + p_e} \right)$ and $\left(\frac{p_e}{p_y + p_e} \right)$. If $\gamma = \frac{p_e}{p_y + p_e}$ and $(1 - \gamma) = \frac{p_y}{p_y + p_e}$, equation (2) can be re-written as:

$$EAO_t = (p_y + p_e) * ((1 - \gamma) * Y_t - \gamma * (E_t)) \quad (3)$$

Where the term $((1 - \gamma) * Y_t - \gamma * (E_t))$ captures the relative importance of the traditional and environmental externalities. Equation (3) allows us to recover the value of the environmentally adjusted output at time t . This can be slightly modified to recover a value of environmentally adjusted output indexed at time t (for simplicity let us assume time 0).

Defining:

$$YI_t = \frac{Y_t}{Y_0} \text{ and } EI_t = \frac{E_t}{E_0}$$

Then an EAO at time t can be rewritten as:

$$EAO_t = p_y * (Y_0 * YI_t) - p_e * (E_0 * EI_t) \quad (4)$$

The EAO can be divided by the value of the environmentally adjusted output at time 0 to obtain an EAO index.

$$EAOI_t = \frac{1}{p_y Y_0 - p_e E_0} (p_y * (Y_0 * YI_t) - p_e * (E_0 * EI_t)) \quad (5)$$

Equation (5) can be rewritten as:

$$EAOI_t = \frac{p_y Y_0 + p_e E_0}{p_y Y_0 - p_e E_0} \left(\frac{p_y Y_0}{p_y Y_0 + p_e E_0} (YI_t) - \frac{p_e E_0}{p_y Y_0 + p_e E_0} (EI_t) \right) \quad (6)$$

If:

$$W = \frac{p_y Y_0 + p_e E_0}{p_y Y_0 - p_e E_0}; \gamma = \frac{p_e E_0}{p_y Y_0 + p_e E_0} \text{ and } 1 - \gamma = \frac{p_y Y_0}{p_y Y_0 + p_e E_0}$$

Equation (6) can be rewritten as:

$$EAOI_t = W * ((1 - \gamma) * YI_t - \gamma * EI_t) \quad (7)$$

Where W is a scaling factor, and the relative weights given to the traditional outputs and environmental externalities are provided by their relative weight in the total absolute combined economic value of output and externalities. In EATFP metrics with an output adjustment, the EATFP index can therefore be expressed as:

$$EATFPO_t = \frac{EAOI_t}{XI_t} = \frac{W * ((1 - \gamma) * YI_t - \gamma * EI_t)}{XI_t} \quad (8)$$

This is important to highlight that, while value of γ is not explicit in EATFPO metrics as it is in ESPI (where $\gamma = \alpha$), the metric can nevertheless be expressed in a similar way.

EATFP – Input adjustment

In the case of an input adjustment, a similar case can be made. Specifically, in the case of one input and constant prices:

X_t - Quantity of input X

XI_t - Input index

p_x - Time-invariant unit price of input X

In the case of input-adjusted EATFP metrics, the adjusted denominator is typically given by an expression of the form:

$$EAI_t = p_x X_t + p_e E_t \quad (9)$$

This expression can be rewritten as:

$$EAI_t = (p_x + p_e) \left(\frac{p_x}{p_x + p_e} X_t + \frac{p_e}{p_x + p_e} E_t \right) \quad (10)$$

If:

$$XI_t = \frac{X_t}{X_0} \text{ and } EI_t = \frac{E_t}{E_0}$$

The environmentally-adjusted input can be re-written as:

$$EAI_t = p_x * (X_0 * XI_t) + p_e * (E_0 * EI_t) \quad (11)$$

This value can be indexed by dividing the expression in equation (11) by EAI_0 . This gives us:

$$EAII_t = \frac{1}{p_x X_0 + p_e E_0} * (p_x * (X_0 * XI_t) + p_e * (E_0 * EI_t)) \quad (12)$$

Where equation (12) can be re-written as :

$$EAII_t = \left(\frac{p_x X_0}{p_x X_0 + p_e E_0} * XI_t + \frac{p_e E_0}{p_x X_0 + p_e E_0} * EI_t \right) \quad (13)$$

If $(1 - \gamma) = \frac{p_x X_0}{p_x X_0 + p_e E_0}$ and $\gamma = \frac{p_e E_0}{p_x X_0 + p_e E_0}$, then equation (13) can be re-written as:

$$EAII_t = (1 - \gamma) XI_t + \gamma * EI_t \quad (14)$$

Which means an EATFP index adjusted through the input side can be expressed as:

$$EATFPI_t = \frac{YI_t}{EAII_t} = \frac{Y_t}{(1 - \gamma) XI_t + \gamma * EI_t} \quad (15)$$

Where gamma can be interpreted as the weight of the externality in the total value of inputs used.

WEP indicator

In the case of WEP, the weight given to the indicator is simply the weight given to the environmental dimensions and it is chosen by the user. Using the notation $g(.)$ to denote a growth rate, WEP is simply:

$$g(WEP) = (1 - \alpha) * g(TFP) - \alpha * g(E) \quad (16)$$

In other words, in the case of WEP, $\gamma = \alpha$. As such:

$$g(WEP) = (1 - \gamma) * g(TFP) - \gamma * g(E) \quad (16)$$

One important point, however, is that in WEP the interpretation of gamma differs somewhat. Whereas in EATFP measures gamma is used to adjust the output or input in levels, in the case of WEP the adjustment is done directly through the growth rates.

Summary

Importantly, with fixed weights and/or prices, all aggregation methods can be expressed as a function of a parameter that captures the relative weight of the environment. The table below summarises the expressions:

Table A A.1. Indicators as a function of gamma

Metric	Expression
EATFPO	$\frac{EAOI_t}{XI_t} = \frac{W * ((1 - \gamma) * YI_t - \gamma * EI_t)}{XI_t}$ $\text{with } W = \frac{p_y Y_0 + p_e E_0}{p_y Y_0 - p_e E_0}; \gamma = \frac{p_e E_0}{p_y Y_0 + p_e E_0}$
EATFPI	$\frac{YI_t}{EAII_t} = \frac{Y_t}{(1 - \gamma)XI_t + \gamma * EI_t}$ $\text{with } \gamma = \frac{p_e E_0}{p_x X_0 + p_e E_0}$
ESPIO	$\frac{(YI_t)^{1-\gamma}(EI_t)^{-\gamma}}{XI_t}$ With $\gamma = \alpha$ and $0 \leq \gamma \leq 1$
ESPII	$\frac{YI_t}{(XI_t)^{1-\gamma}(EI_t)^{\gamma}}$ With $\gamma = \alpha$ and $0 \leq \gamma \leq 1$
WEP	$g(WEP) = (1 - \gamma) * g(TFP) - \gamma * g(EI)$ With $\gamma = \alpha$ and $0 \leq \gamma \leq 1$

Annex B. Derivations of the necessary conditions for SPG to exceed TFP growth

When is SPG positive and how does it relate to TFP?

Using the previously defined nomenclature:

YI_t - Index of agricultural output (excluding aquaculture) at time t

XI_t - Input index at time t

EI_t - Index of the environmental externalities (for the time being, let's assume only one for ease of exposition)

The TFP index at time t is given by:

$$TFP_t = \frac{YI_t}{XI_t} \quad (18)$$

It therefore follows that the growth rate of TFP is given by:

$$\ln\left(\frac{TFP_t}{TFP_{t-1}}\right) = \ln\left(\frac{YI_t}{YI_{t-1}}\right) - \ln\left(\frac{XI_t}{XI_{t-1}}\right) \quad (19)$$

In other words, TFP growth will be positive when growth rate of outputs exceeds the growth rate of inputs. In other words, TFP growth is positive when there is at least relative decoupling between output and inputs (i.e. when more is produced with less).

EATFP – Output adjustment

As will be seen below, similar expressions can be derived for environmentally adjusted metrics. Taking logs this can be expressed as:

$$\ln(EATFPO_t) = \ln(EAOI_t) - \ln(XI_t) \quad (20)$$

And the growth rate can be given by:

$$\ln\left(\frac{EATFPO_t}{EATFPO_{t-1}}\right) = \ln\left(\frac{EAOI_t}{EAOI_{t-1}}\right) - \ln\left(\frac{XI_t}{XI_{t-1}}\right) \quad (21)$$

What this expression means is that SPG will be positive whenever the rate of growth of the environmentally adjusted output exceeds that of the traditional inputs. By combining equation (19) with equation (21), (21) is larger than (19) when:

$$\ln\left(\frac{EAOI_t}{EAOI_{t-1}}\right) > \ln\left(\frac{YI_t}{YI_{t-1}}\right) \quad (22)$$

Since $EAOI_t = W * ((1 - \gamma) * YI_t - \gamma * EI_t)$, it follows that for condition (22) to hold growth rate of the environmentally adjusted output needs to exceed that of traditional output. This occurs when:

$$\frac{W * ((1 - \gamma) * YI_t - \gamma * EI_t) - W * ((1 - \gamma) * YI_{t-1} - \gamma * EI_{t-1})}{W * ((1 - \gamma) * YI_{t-1} - \gamma * EI_{t-1})} > \frac{YI_t - YI_{t-1}}{Y_{t-1}} \quad (23)$$

Rearranging (23) :

$$\frac{((1 - \gamma) * (YI_t - YI_{t-1}) - \gamma * (EI_t - EI_{t-1}))}{((1 - \gamma) * YI_{t-1} - \gamma * EI_{t-1})} > \frac{YI_t - YI_{t-1}}{Y_{t-1}} \quad (24)$$

Defining $YI_t - YI_{t-1}$ as ΔYI and $EI_t - EI_{t-1}$ as ΔEI , the inequality can be expressed as:

$$((1 - \gamma)\Delta YI - \gamma\Delta EI) * Y_{t-1} > \Delta YI * ((1 - \gamma) * YI_{t-1} - \gamma * EI_{t-1}) \quad (25)$$

Expanding both sides:

$$(1 - \gamma)\Delta YI * YI_{t-1} - \gamma\Delta EI * Y_{t-1} > (1 - \gamma) * YI_{t-1} * \Delta YI - \gamma * EI_{t-1} * \Delta YI \quad (26)$$

Subtracting both sides implies that:

$$-\gamma * \Delta EI * Y_{t-1} > -\gamma * EI_{t-1} * \Delta YI \quad (27)$$

Which can be rearranged as:

$$\frac{\Delta YI}{Y_{t-1}} > \frac{\Delta EI}{EI_{t-1}} \quad (28)$$

What this implies is that condition (22) holds (i.e. the growth rate of SPG exceeds TFP) when the growth rate of traditional output exceeds that of the externality.

EATFP – Input adjustment

For EATFP with an input adjustment, taking logs implies that:

$$\ln(\text{EATFPI}_t) = \ln(YI_t) - \ln(EAII_t) \quad (29)$$

And the growth rate can be given by:

$$\ln\left(\frac{\text{EATFPI}_t}{\text{EATFPI}_{t-1}}\right) = \ln\left(\frac{YI_t}{YI_{t-1}}\right) - \ln\left(\frac{EAII_t}{EAII_{t-1}}\right) \quad (30)$$

What this expression means is that SPG will be positive whenever the rate of growth of outputs exceeds that of the environmentally-adjusted inputs. By combining equation (19) with equation (30), (30) is larger than (19) when:

$$\ln\left(\frac{EAII_t}{EAII_{t-1}}\right) < \ln\left(\frac{XI_t}{XI_{t-1}}\right) \quad (31)$$

Since $EAII_t = (1 - \gamma)XI_t + \gamma * EI_t$, it follows that the following inequality needs to hold:

$$\frac{(1 - \gamma)XI_t + \gamma * EI_t - ((1 - \gamma)XI_{t-1} + \gamma * EI_{t-1})}{(1 - \gamma)XI_{t-1} + \gamma * EI_{t-1}} < \frac{XI_t - XI_{t-1}}{XI_{t-1}} \quad (32)$$

Rearranging (32):

$$\frac{(1 - \gamma) * (XI_t - XI_{t-1}) + \gamma(EI_t - EI_{t-1})}{(1 - \gamma)XI_{t-1} + \gamma * EI_{t-1}} < \frac{XI_t - XI_{t-1}}{XI_{t-1}} \quad (33)$$

Defining $XI_t - XI_{t-1}$ as ΔXI and $EI_t - EI_{t-1}$ as ΔEI =:

$$\frac{(1 - \gamma) * (\Delta XI) + \gamma(\Delta EI)}{(1 - \gamma)XI_{t-1} + \gamma * EI_{t-1}} < \frac{\Delta XI}{XI_{t-1}} \quad (34)$$

Expanding both sides implies that:

$$(1 - \gamma)\Delta XI * XI_{t-1} + \gamma\Delta EI * XI_{t-1} < (1 - \gamma) * XI_{t-1} * \Delta XI + \gamma * EI_{t-1} * \Delta XI \quad (35)$$

Which simplifies to:

$$\gamma\Delta EI * XI_{t-1} < \gamma * EI_{t-1} * \Delta XI \quad (36)$$

Which simplifies further to:

$$\frac{\Delta EI}{EI_{t-1}} < \frac{\Delta XI}{XI_{t-1}} \quad (37)$$

Which essentially means that the growth rate of SPG exceeds the growth rate of TFP when the growth rate of traditional output exceeds that of the externalities.

WEP

$$(1 - \gamma) * \frac{(TFP_t - TFP_{t-1})}{TFP_{t-1}} - \gamma * \frac{EI_t - EI_{t-1}}{EI_{t-1}} > \frac{(TFP_t - TFP_{t-1})}{TFP_{t-1}} \quad (38)$$

This can be re-arranged as:

$$\gamma \left(\frac{\Delta EI}{EI_{t-1}} + \frac{\Delta TFP}{TFP_{t-1}} \right) < 0 \quad (39)$$

For $\gamma > 0$, this that the term inside the brackets needs to be negative, which simplifies to:

$$\frac{\Delta EI}{EI_{t-1}} < -\frac{\Delta TFP}{TFP_{t-1}} \quad (40)$$

In other words, this implies that in the case of WEP, for growth rates of WEP to be higher than those of TFP, the rate of abatement of externalities needs to exceed the growth rate of TFP.

ESPI – Input adjustment

For ESPI with an input adjustment, taking logs leads to the following equation:

$$\ln(ESPI_t) = \ln(YI_t) - (1 - \gamma) * \ln(XI_t) - \gamma * \ln(EI_t) \quad (41)$$

And the growth rate can be given by:

$$\ln\left(\frac{ESPI_t}{ESPI_{t-1}}\right) = \ln\left(\frac{YI_t}{YI_{t-1}}\right) - (1 - \gamma) * \ln\left(\frac{XI_t}{XI_{t-1}}\right) - \gamma * \ln\left(\frac{EI_t}{EI_{t-1}}\right) \quad (42)$$

What this expression means is that SPG will be positive whenever the rate of growth of outputs exceeds that of the environmentally adjusted inputs. By combining equation (19) with equation (42), (42) is larger than (19) when:

$$(1 - \gamma) * \ln\left(\frac{XI_t}{XI_{t-1}}\right) + \gamma * \ln\left(\frac{EI_t}{EI_{t-1}}\right) < \ln\left(\frac{XI_t}{XI_{t-1}}\right) \quad (43)$$

Which simplifies to:

$$\gamma * \ln\left(\frac{EI_t}{EI_{t-1}}\right) < \gamma * \ln\left(\frac{XI_t}{XI_{t-1}}\right) \quad (44)$$

Which is equivalent to:

$$\frac{\Delta EI}{EI_{t-1}} < \frac{\Delta XI}{XI_{t-1}} \quad (45)$$

Which essentially means that the growth rate of SPG exceeds the growth rate of TFP when the growth rate of traditional output exceeds that of the externalities.

ESPI – Output adjustment

For ESPIO with an input adjustment, taking logs implies that:

$$\ln(\text{ESPIO}_t) = (1 - \alpha) \ln(YI_t) - \ln(XI_t) - \alpha \ln(EI_t) \quad (46)$$

And the growth rate can be given by:

$$\ln\left(\frac{\text{ESPIO}_t}{\text{ESPIO}_{t-1}}\right) = (1 - \gamma) * \ln\left(\frac{YI_t}{YI_{t-1}}\right) - \ln\left(\frac{XI_t}{XI_{t-1}}\right) - \gamma * \ln\left(\frac{EI_t}{EI_{t-1}}\right) \quad (47)$$

Combining equation (47) with equation (19) implies that ESPIO growth rates are higher than TFP growth rates when:

$$(1 - \gamma) * \ln\left(\frac{YI_t}{YI_{t-1}}\right) - \gamma * \ln\left(\frac{EI_t}{EI_{t-1}}\right) > \ln\left(\frac{YI_t}{YI_{t-1}}\right) \quad (48)$$

Which further simplifies to:

$$\gamma * \left(\ln\left(\frac{EI_t}{EI_{t-1}}\right) + \ln\left(\frac{YI_t}{YI_{t-1}}\right)\right) < 0 \quad (49)$$

For a positive value of γ this requires:

$$\ln\left(\frac{EI_t}{EI_{t-1}}\right) < -\ln\left(\frac{YI_t}{YI_{t-1}}\right) \quad (50)$$

In other words, a similar condition to that required by WEP applies. The growth rate of the environmentally adjusted metric will exceed that of traditional TFP when the speed of abatement of the externality exceeds the speed of output growth.

Summary

Metric	SPG>0 when	SPG>TFP when
EATFPO	$\ln\left(\frac{(1-\gamma) * YI_t - \gamma * EI_t}{(1-\gamma) YI_{t-1} - \gamma * EI_{t-1}}\right) > \ln\left(\frac{XI_t}{XI_{t-1}}\right)$	$\frac{\Delta EI}{EI_{t-1}} < \frac{\Delta YI}{Y_{t-1}}$
EATFPI	$\ln\left(\frac{YI_t}{YI_{t-1}}\right) > \ln\left(\frac{(1-\gamma)XI_t + \gamma * EI_t}{(1-\gamma)XI_{t-1} + \gamma * EI_{t-1}}\right)$	$\frac{\Delta EI}{EI_{t-1}} < \frac{\Delta XI}{XI_{t-1}}$
ESPIO	$(1-\gamma) \ln\left(\frac{YI_t}{YI_{t-1}}\right) - \gamma \ln\left(\frac{EI_t}{EI_{t-1}}\right) > \ln\left(\frac{XI_t}{XI_{t-1}}\right)$	$\frac{\Delta EI}{EI_{t-1}} < -\frac{\Delta YI}{Y_{t-1}}$
ESPII	$\ln\left(\frac{YI_t}{YI_{t-1}}\right) > (1-\gamma) \ln\left(\frac{XI_t}{XI_{t-1}}\right) + \gamma \ln\left(\frac{EI_t}{EI_{t-1}}\right)$	$\frac{\Delta EI}{EI_{t-1}} < \frac{\Delta XI}{XI_{t-1}}$
WEP	$(1-\gamma) * \ln\left(\frac{TFP_t}{TFP_{t-1}}\right) > \gamma * \ln\left(\frac{EI_t}{EI_{t-1}}\right)$	$\frac{\Delta EI}{EI_{t-1}} < -\frac{\Delta TFP}{TFP_{t-1}}$

Annex C. Data and methods

Data series and explanation of constructed variables

For the three countries included in the study the following data sources are common across all countries.

Table A C.1. Data sources used in the analysis

Variable	Definition	Source
Output	Gross Value of Production (constant 2014-2016 international USD)	FAOSTAT
Input index	Agricultural input index is an aggregation of the quantity of land, labour, capital, and two representative intermediate inputs (fertilizer and feed). Aggregation of the different inputs is carried out using representative input cost shares that estimate the production elasticities of different inputs for specific countries/regions. Cost shares are compiled from multiple empirical analyses multiple sources and details are provided in Table A2 of Fuglie (2015) (Fuglie, 2015 ^[89]) .	USDA – International Agricultural Productivity database In order to construct the variables in the database, different data sources are used. Data on outputs for crops, livestock, and aquaculture are sourced from FAO. Data on land, capital, fertilizer and animal feed inputs are also mainly sourced from FAO, whereas agricultural labour data is sourced from ILO. Data on cost shares are based on empirical studies.
Cost shares	Weights used to aggregate importance of the input in the input index	USDA – International Agricultural Productivity database
Ammonia emissions	NH3 emissions from agriculture (Tonnes)	Mexico: OECD Agri-environmental indicators database; Measure: Agricultural ammonia (NH3) from 2005-2018. For pre-2005 data, the 2005 value for ammonia emissions is adjusted using the growth rates from the EDGAR database. For 2019, the 2018 value is adjusted using the growth rates implied by the EDGAR database ammonia series. Netherlands: OECD Agri-environmental indicators database; Measure: Agricultural ammonia (NH3) New Zealand: JRC - Emissions Database for Global Atmospheric Research (EDGAR)
Nitrogen surplus, Phosphorus surplus	Nitrogen/Phosphorus inputs minus (excess of nutrient inputs vs. outputs)	OECD Agri-environmental indicators database: Nutrient balances Measure- Balance (inputs minus outputs)
GHG emissions from agriculture	GHG emissions (CO ₂ , N ₂ O and CH ₄) in CO ₂ equivalent	OECD Agri-environmental indicators database ; Measure : Total greenhouse gas emissions from agriculture (Tonnes of CO ₂ -equivalent, thousands)

In the case of nutrient balances, since the shadow values used in the analysis are generally based on kg of nutrient removed from water, there is an attempt to adjust the overall nutrient balances to capture the average proportion of the nutrient surplus that reaches water bodies (e.g. leaching and run-off). This is done in the following way. First, a country-specific estimate (whenever data was available) of the fraction of leaching and run-off is obtained. This can either be a fraction of total nutrient inputs or, alternatively, an estimate of the absolute levels of leaching and run-off.

When a fraction is used, it is assumed to be time-invariant. Using this fraction the total quantity of nutrient leaching and run-off is obtained over the entire analysed period (by summing the annual estimated quantities of leaching and run-off) and the ratio of this value to the total nutrient balance is used to obtain the average share of leaching and run-off in the nutrient balance. In the cases where only an absolute value of leaching and run-off per hectare is available, this value is compared to the average nutrient balance per hectare over the analysed time period to derive an average fraction of leaching and run-off as a share of the nutrient balance.

The Table below summarises the data sources used to estimate the quantity of leaching and run-off.

Table A C.2. Assumptions – Leaching and run-off

Country	Definition	Explanation	Source
New Zealand	Estimated fraction of run-off and leaching (nitrogen)	A share of 10% nitrogen inputs is used for crops and 7% of total nitrogen inputs is used for pasture. Putting the two together a value of approximately 8% is obtained.	National Inventory Report of New Zealand 2024 (Ministry for the Environment, 2025 ^[70])
New Zealand	Estimated fraction of run-off and leaching (phosphorus)	A value of 1.5 kg/ha/yr is used as it is the mid-point provided in a paper focusing specifically on New Zealand. This amounts to approximately 16% of the surplus.	Selbie et al. (2013) - Understanding the distribution and fate of nitrogen and phosphorus in OVERSEER (Selbie et al., 2013 ^[90])
Netherlands	Estimated fraction of run-off and leaching (nitrogen)	A value of 14% is used. This is the mid-point of the range of values used in the National Inventory Report (ranges from 13-15%)	Van der Zee et al. 2025 - Methodology for the calculation of emissions from agriculture (RIVM, 2025 ^[91])
Netherlands	Estimated fraction of run-off and leaching (Phosphorus)	A modelling assumption of 10% of phosphorus inputs was used. This is a global average, but is used in national modelling approaches (e.g. ReCiPe)	Bouwman et al. (2009) (Bouwman, Beusen and Billen, 2009 ^[92] ; RIVM, 2016 ^[93])
Mexico	Estimated fraction of run-off and leaching (nitrogen)	A value of 24% is used, as used by Mexico in its National Inventory. This is a default value.	(Gobierno de México, 2025 ^[50]) 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories – Chapter 11 (Hergoualch et al., 2019 ^[94])
Mexico	Estimated fraction of run-off and leaching (Phosphorus)	A modelling assumption of 10% of phosphorus inputs was used. This is a global average.	(Bouwman, Beusen and Billen, 2009 ^[92])

Prices used for the environmental externalities

Abatement costs and societal costs of environmental externalities are associated with high levels of uncertainty. The Table below highlights the prices that were used per unit of externality as well as their respective source. The prices presented in the Table below were then converted into 2014-2016 international USD. It is important to notice, however, that there is substantial uncertainty around estimates of unit costs for environmental valuation. The Table below highlights the choices that were made for this analysis and the rationale for the decisions.

Table A C.3. Prices environmental externalities

Country	Environmental dimension	Perspective	Value used	Explanation	Source
New Zealand	Ammonia	Producer	USD 1 500/tonne	These are based on abatement costs in a study focusing on Australia as no costs specific to New Zealand were found. In the study for Australia, the ranges of costs for abatement options from a meta-analysis are very large, with positive abatement costs ranging from USD 60/tonne to USD 9 390/tonne. A value of USD 1 500 was selected for two reasons. First, it falls within the range of USD 500-USD 4 000 which is the range where most abatement options lie. Second, it is consistent with estimates of global abatement costs.	Zhang et al. (2022) – Table S8
Netherlands	Ammonia	Producer	USD 1 900/tonne	The value is obtained based on a country-specific range of abatement values provided in Zhang et al. (2021). For the Netherlands, a range of USD 300-USD 3 500 per tonne is provided and the mid-point of this range is selected.	Zhang et al. (2021) – Table 1;
Mexico	Ammonia	Producer	USD 1 500/tonne	No country-specific abatement cost were found for Mexico and this motivated the use for a global abatement cost.	Gu et al. (2021)
New Zealand	Ammonia	Societal	USD 10 790/tonne (constant 2020 USD)	The estimated unit cost from NH3 emissions to Air from the database SPIQ-FS modelling used for State of Food and Agriculture 2023 report is used	Lord (2023)
Netherlands	Ammonia	Societal	EUR 28 700/tonne (2021 constant EUR)	This is the central value provided by the Environmental Prices Handbook 2024: EU27 version. Note: This price is similar to the value in the SPIQ-FS model (31 000 constant 2020 USD/tonne)	De Vries et al. (2024) version 1.0
Mexico	Ammonia	Societal	USD 2 700/tonne (constant 2020 USD)	The estimated unit cost from NH3 emissions to Air from the database SPIQ-FS modelling used for State of Food and Agriculture 2023 report is used.	Lord (2023)
New Zealand	GHG	Producer	EUR 30.6 /tonne (constant 2023 EUR)	This is the estimated effective carbon rate for New Zealand for 2023.	OECD effective Carbon rates database
Netherlands	GHG	Producer	EUR 34.59 /tonne (constant 2023 EUR)	This is the ETS component of the estimated effective carbon rate for the Netherlands for 2023.	OECD effective Carbon rates database
Mexico	GHG	Producer	EUR 1.58 /tonne (constant 2023 EUR)	This is the carbon tax component of the estimated effective carbon rate for Mexico for 2023.	OECD effective Carbon rates database
New Zealand	GHG	Societal	USD 51/tonne (constant 2020 USD)	The estimated unit cost for CO ₂ emissions (including marginal damage costs attributed to mortality) from the database SPIQ-FS modelling used for State of Food and Agriculture 2023 report is used	Lord (2023)

Country	Environmental dimension	Perspective	Value used	Explanation	Source
Netherlands	GHG	Societal	USD 51/tonne (constant 2020 USD)	The estimated unit cost for CO ₂ emissions (including marginal damage costs attributed to mortality) from the database SPIQ-FS modelling used for State of Food and Agriculture 2023 report is used	Lord (2023)
Mexico	GHG	Societal	USD 51/tonne (constant 2020 USD)	The estimated unit cost for CO ₂ emissions (including marginal damage costs attributed to mortality) from the database SPIQ-FS modelling used for State of Food and Agriculture 2023 report is used	Lord (2023)
New Zealand	Phosphorus	Producer	USD 4 500/tonne	Estimated ranges differ substantially in the study found for New Zealand, with cost ranges between 0-200. However, several abatement options had a lower-bound range of 2-5 dollars per kg. A value of USD 4 500 was selected.	McDowell and Nash (2012)
Netherlands	Phosphorus	Producer	EUR 3 740/tonne (constant 2021 EUR)	This is the central value in the Environmental Prices Handbook 2024: EU27 version 1. While this handbook generally focuses on valuations from a more societal perspective (e.g. costs of water treatment). In the Netherlands, this value was used as, as noted by van Grinsven et al. (2021), measures to improve water treatment would be attractive because these are generally more cost-effective than measures to reduce nutrient inputs from agricultural fields	De Vries et al. (2024) version 1.0
Mexico	Phosphorus	Producer	EUR 3 631/tonne (constant 2021 EUR)	No country-specific estimates of abatement costs for Phosphorus were found for Mexico. A central value of Environmental Prices Handbook 2024: EU27 version 1 for phosphorus was used and the value for the EU was adjusted by the ratio of general water treatment costs in Mexico and the Netherlands.	Authors' calculations, based on De Vries et al. (2024) version 1.0
New Zealand	Phosphorus	Societal	NZD 8 000/tonne (approximately USD 4 683)	The 2025/26 Kapiti Coast water treatment charges for Phosphorus are used. Estimates from Kapiti coast were used as data was publicly available and expressed in tonnes of nutrient. To improve relevance of the unit cost used in the future, it would be best to obtain similar valuations from areas accounting for a larger share of agricultural production.	Kapiti coast water and wastewater treatment charges
Netherlands	Phosphorus	Societal	EUR 3 740/tonne (constant 2021 EUR)	This is the central value in the Environmental Prices Handbook 2024: EU27 version 1. While this handbook generally focuses on valuations from a more societal perspective (e.g. costs of water treatment). In the Netherlands, this value was used as, as noted by van Grinsven et al. (2021), measures to improve water treatment would be attractive because these are generally more cost-effective than measures to reduce nutrient inputs from agricultural fields	De Vries et al. (2024) version 1.0

Country	Environmental dimension	Perspective	Value used	Explanation	Source
Mexico	Phosphorus	Societal	EUR 3 631 /tonne (constant 2021 EUR)	No country-specific estimates of abatement costs for Phosphorus for Mexico were found. The value used therefore relies on the central value of Environmental Prices Handbook 2024: EU27 version 1 for phosphorus, adjusting this value by the ratio of general water treatment costs between Mexico and the Netherlands.	Authors' calculations, based on De Vries et al. (2024) version 1.0
New Zealand	Nitrogen	Producer	NZD 1 900/tonne	The lower range provided in Vibart et al. (2015) was used, noting that the range is very wide, ranging from NZD 1.91/tonne to over NZD 34/tonne	Vibart et al. (2015)
Netherlands	Nitrogen	Producer	EUR 2 600/tonne	This is the value is used in a previous valuation study of environmental externalities for the Netherlands by Jongeneel et al. (2016)	Jongeneel et al. (2016)
Mexico	Nitrogen	Producer	USD 700	No country-specific abatement cost was found for nitrogen for Mexico. The study therefore relies on a global study that estimates implementation cost of cost-effective mitigation of nitrogen pollution from global croplands. To derive the unit value, the estimated implementation cost of measures for Mexico are divided by the modelled changes in leaching and run-off	Authors' calculations, based on Gu et al. (2023)
New Zealand	Nitrogen	Societal	NZD 2 750/tonne	The 2025/26 Kapiti Coast water treatment charges for Nitrogen are used	Kapiti coast water and wastewater treatment charges
Netherlands	Nitrogen	Societal	EUR 4 300/tonne (constant 2021 EUR)	This is the central value in the Environmental Prices Handbook 2024: EU27 version 1 for a tonne of nitrogen emission to freshwater	De Vries et al. (2024) version 1.0
Mexico	Phosphorus	Societal	EUR 4 175/tonne (constant 2021 EUR)	No country-specific estimates of abatement costs for Nitrogen for Mexico were found. Central value of Environmental Prices Handbook 2024: EU27 version 1 were used and these values were adjusted by the ratio of general water treatment costs in Mexico and the Netherlands	Authors' calculations, based on De Vries et al. (2024) version 1.0

Constructing the indicators

When constructing the different indicators, the paper tries to remain as close to the USDA indices as possible so as to allow for greater comparability between the estimated growth rates and the growth rates from the USDA IAP data product (U.S. Department of Agriculture, 2024^[37]).

One important difference is that with regards to the output index is that in this paper aquaculture is not included in the output index, though this does not change the value of production much for the three country case studies where aquaculture is generally a low share of total output.

With regards to the adjusted indicators, the Table below summarises the approach used.

Table A C.4. Calculation of the adjusted indicators

Indicator	Variables used
EATFPO	The Environmentally adjusted output is derived by subtracting the monetary value of externalities (i.e. prices in Table A C.3. in 2015 USD by quantities of the externalities). The denominator of the EATFPO index is the input index of the USDA IAP dataset.
EATFPI	The output index is based on the Gross Value of Production (constant 2014-2016 international USD) from FAOSTAT as in the USDA IAP database. The main difference is that aquaculture is excluded. For the denominator, since the actual monetary value of inputs are not observed (the disaggregated quantities by input, however, are and the relative importance of different inputs are captured by cost shares) in the USDA IAP database, a simplifying assumption is required in order to have an approximation of the monetary value of inputs. Specifically, it is assumed that, for 2015, the value of output is equal to the value of inputs, and this assumption provides an assumed monetary value of inputs for a base year. The USDA input index is then use to change the monetary estimate over time. To construct the environmentally-adjusted input variable the monetary value of the externalities is added to the estimated value of the traditional inputs. We note that the USDA ERS IAP dataset includes data on input quantities by type of input and includes different types of outputs (e.g. disaggregated by crops, livestock and aquaculture). We note that in certain contexts (e.g. several EU countries), there are cases where input costs are higher than output values. In such contexts, the simplifying assumption would result in a higher weight of the environment than if the true market value of inputs was actually observed.
ESPIO	The environmentally adjusted output is constructed using the formula highlighted in Cobourn et al. (2024). The value of alpha in this paper is defined as the share of the monetary value of the externality as a proportion of the total joint monetary value of the externality and output in 1990. One difference compared to Cobourn et al. (2024) is that instead of benchmarking quantities against the OECD average in 1990, in this paper 2015 is used as a benchmark period. $\alpha o = \frac{p_e E_{1990}}{VoP_{1990} + p_e E_{1990}}$ The input index is constructed by constructing quantity indices for each input against OECD average values for 2015 and using the USDA cost shares.
ESPII	The numerator is simply the output index (benchmarked against the 2015 OECD average value). The input index is constructed by assigning a weight to traditional inputs and to the environmental externalities. The weight assigned to the environmental externalities is given by the ratio of the monetary value of the externality as a share of the total monetary value of the externality plus the estimated value of inputs in 1990. $\alpha i = \frac{p_e E_{1990}}{VoI_{1990} + p_e E_{1990}}$
WEPO	This is calculated as the weighted average of TFP growth rate and the growth rate of externalities. The weight assigned to the externalities is the same parameter used in the ESPIO indicator.
WEPI	This is calculated as the weighted average of TFP growth rate and the growth rate of externalities. The weight assigned to the externalities is the same parameter used in the ESPII indicator

Based on the above, the values of alpha used for ESPII, ESPIO, WEPI, WEPO are the following:

Table A C.5. Parameters used as environmental weights in ESPII, ESPIO, WEPI and WEPO

Adjustment	Country	Externality(ies)	Producer perspective	Society's perspective
Input	NZ	GHG only	0.09	0.16
Input	NZ	Ammonia	0.01	0.10
Input	NZ	Nitrogen	0.02	0.02
Input	NZ	Phosphorus	0.01	0.01
Input	NZ	GHG, Nitrogen + Phosphorus	0.12	0.18
Input	NZ	GHG, Nitrogen, Phosphorus + Ammonia	0.13	0.25
Output	NZ	GHG only	0.11	0.18
Output	NZ	Ammonia	0.01	0.11
Output	NZ	Nitrogen	0.02	0.02
Output	NZ	Phosphorus	0.01	0.01
Output	NZ	GHG, Nitrogen + Phosphorus	0.13	0.20
Output	NZ	GHG, Nitrogen, Phosphorus + Ammonia	0.14	0.28
Input	MEX	GHG only	0.00	0.12
Input	MEX	Ammonia	0.02	0.04
Input	MEX	Nitrogen	0.03	0.14
Input	MEX	Phosphorus	0.01	0.01
Input	MEX	GHG, Nitrogen + Phosphorus	0.03	0.23
Input	MEX	GHG, Nitrogen, Phosphorus + Ammonia	0.05	0.25
Output	MEX	GHG only	0.01	0.18
Output	MEX	Ammonia	0.03	0.06
Output	MEX	Nitrogen	0.04	0.21
Output	MEX	Phosphorus	0.01	0.01
Output	MEX	GHG, Nitrogen + Phosphorus	0.06	0.33
Output	MEX	GHG, Nitrogen, Phosphorus + Ammonia	0.08	0.36
Input	NLD	GHG only	0.04	0.07
Input	NLD	Ammonia	0.04	0.39
Input	NLD	Nitrogen	0.02	0.04
Input	NLD	Phosphorus	0.00	0.00
Input	NLD	GHG, Nitrogen + Phosphorus	0.07	0.11
Input	NLD	GHG, Nitrogen, Phosphorus + Ammonia	0.10	0.43
Output	NLD	GHG only	0.05	0.08
Output	NLD	Ammonia	0.04	0.42
Output	NLD	Nitrogen	0.03	0.05
Output	NLD	Phosphorus	0.00	0.00
Output	NLD	GHG, Nitrogen + Phosphorus	0.08	0.12
Output	NLD	GHG, Nitrogen, Phosphorus + Ammonia	0.11	0.46

Decomposing the results

Using a log decomposition of EATFPI can be decomposed as:

$$\ln\left(\frac{EATFPI_t}{EATFPI_{t-1}}\right) = \ln\left(\frac{Y_t}{Y_{t-1}}\right) - \ln\left(\frac{EAI_t}{EAI_{t-1}}\right)$$

Where EAI is the value of environmentally adjusted input (not the index). The contribution of output is given simply by the (log) growth rate of output. For the environmentally adjusted input, it needs to be separated further into the contribution of inputs and externalities. The contribution of the externalities can be calculated in the following way:

$$Cont\ Externalities_{t,t-1} = \left(\frac{1}{2}\left(\frac{p_e E_t}{p_x X_t + p_e E_t}\right) + \frac{1}{2}\left(\frac{p_e E_{t-1}}{p_x X_{t-1} + p_e E_{t-1}}\right)\right) * \ln\left(\frac{p_e E_t}{p_e E_{t-1}}\right)$$

Which is essentially the average weight of the externalities as a share of the total value of inputs plus externalities times the growth rate in the value of externalities. Note that for the contribution of a single externality, the calculation is essentially the same. The only difference is that the growth rate of that externality is multiplied by the ratio of the monetary value of that externality as a share of the value of Inputs plus externalities.

Similarly, the contribution of traditional inputs between periods t-1 and t is given by:

$$Cont\ inputs_{t,t-1} = \left(\frac{1}{2}\left(1 - \frac{p_e E_t}{p_x X_t + p_e E_t}\right) + \frac{1}{2}\left(1 - \frac{p_e E_{t-1}}{p_x X_{t-1} + p_e E_{t-1}}\right)\right) * \ln\left(\frac{p_x X_t}{p_x X_{t-1}}\right)$$

To produce the decomposition figures the average contributions of the different components is calculated.